TECHNICAL SPECIFICATION · REVERSE-ENGINEERING REPORT

Survey Feedback Compliance Datastore Update

Survicate 24-hour patient surveys → cleaned Excel exports → Power Query → SPC (statistical process control) sheets → six compliance rows in the data store, reverse-engineered from an operator screen recording

Document	Technical Specification — reverse-engineered legacy workflow (as-is) and target architecture (to-be)
Subject workflow	Patient survey feedback compliance (People › Feedback): Survicate exports → *24 hr All Surveys.xlsx* and *Survey Management.xlsm* → *Compliance Data.xlsm* → dashboard
Primary evidence	Meeting with Andrew R. Biggins-20260814_111322-Meeting Recording.mp4 — 1920×1080, 16 fps, 7:07 (428 s); camera off. Recorded first of the 14 August videos, before Training (11:33) and Complications (12:05)
Secondary evidence	…111322….srt (matches this video) and …111322….txt (TurboScribe)
Timestamps	Video positions; every figure matched to a verified frame at 0.25 s steps (fractions shown rounded down)
Companion documents	Feedback_PreBuild_Client_Requirements.pdf (same folder); specs 01–04
Report date	16 October 2026
Status	Draft for business-owner review. Free-text patient comments are blurred in every screenshot

What the recording shows. The operator downloads four 24-hour patient surveys from Survicate, removes “spoilt” responses by hand, saves the files to a shared folder, refreshes a first workbook (24 hr All Surveys) and then a second one (Survey Management) that computes SPC charts and six compliance rows, which are pasted into the data store. “This was designed by Matt, so we just follow his process.”

- **Two overlapping workbooks.** Both have Requirements and Compliance Timeframes, and both build the same 24 hr All Surveys table. The operator says “I’m not even sure we need to complete the first one”. Their compliance type names differ (“SPC Flagged - All” vs “Feedback - SPC Flagged - All”), and the data store holds both sets.
- **The rule is statistical, not a fixed target.** Monthly averages of satisfaction (1–5) and Friends & Family (1–5) per service are tested with SPC rules (outside 3σ limits, 7 points above/below the mean, 7-point trends, 2 of 3 near a limit). A type is compliant when no rule fires for last month: “Satisfaction and F&F within SPC criteria”.
- **Spoken thresholds differ from the logic:** “four and five is acceptable” is said, but the sheets work on monthly averages and SPC limits; for Friends and Family the operator says “less than five is not” acceptable.
- **Fragile ingestion:** exports have hash-like file names, need manual cleaning, and the Self Pay query showed a warning and the combined query an error during refresh (“it’s having trouble finding something”).
- **Hidden cross-tool link:** the Compliance tab reads `[Incident Management.xlsm]Estates` by full path, so this tool depends on another tool being present and current.
- **Usual leftovers:** Every x months 0.5, Missing TRUE, Issue 10, “UNABLE TO REFRESH – linked workbook”, TODAY()-based month selection.

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0. Evidence Base, Method & Reading Guide

0.1 Source artefacts

Artefact	What it is	Used for
...20260814_111322-Meeting Recording.mp4	Teams recording, 7:07, Git-LFS object 39 MB	Screenshots, step timings
...20260814 111322-Meeting Recording.srt / .txt	Timed subtitles and transcript of the same meeting	Timestamps and wording

0.2 Method and redaction

- Frames matched to verified grid frames at 0.25 s steps; cropped to the Excel window, or full screen for browser and Explorer views.
- **Redaction:** the free-text feedback columns (raw export Q#4 onward; 24 hr All Surveys “Feedback”) and the formula bar when it shows a comment are blurred. Scores, dates and service codes are kept. Staff names appear in data-store screenshots, as in specs 01 and 04.

1. Complete File, Workbook & Sheet Inventory

1.1 Files and systems

#	File / system	Role	Evidence
S1	**Survicate** (panel.survicate.com)	Survey platform. Folders: NHS ASD, NHS CTS, NHS VAS & POD, Self-Pay, Staff Feedback, Audits	Figure 2 [00:00:48], Figure 3 [00:01:15]
F1	Four exports (50f982bbbd00319b, 8f3115757b582202, a16d7414468e60c1, 92797e16bc759d36 .xlsx)	Raw survey responses, one per survey, cleaned by hand	Figure 4 [00:01:24], Figure 5 [00:01:36]
F2	**24 hr All Surveys** (.xlsx) in Z:\…\Data Store\Surveys	First tool: Power Query combine + Requirements + Timeframes + SPC sheets	Figure 7 [00:02:30]
F3	**Survey Management.xlsm** in Z:\…\User input	Second tool: same data plus the Compliance output tab	Figure 13 [00:04:27], Figure 6 [00:02:21]
F4	Incident Management.xlsm (Estates tab)	Read by a Survey Management formula	Figure 21 [00:06:17]
F5	**Compliance Data.xlsm**	Data store	Figure 22 [00:06:36]–Figure 24 [00:06:46]

1.2 Tabs

Workbook	Tabs (in order)
24 hr All Surveys	Sheet10 · 24 hr All Surveys · Requirements · Compliance Timeframes · SPC - ND - F&F · SPC - CTS - F&F · …
Survey Management	Compliance · 24 hr All Surveys · Requirements · Compliance Timeframes · SPC - ND/CTS/VAS/Self Pay/NHS/All - F&F · SPC - ND/CTS/VAS/Self Pay/NHS/All - CSAT · Self Pay Types
Power Query (both)	24 hr Self Pay Survey · 24 hr ND Survey · 24 hr VAS Survey · 24 hr CTS Survey · Self_Pay_Types · 24 hr All Surveys = Table.Combine(…)

1.3 Overview screenshots

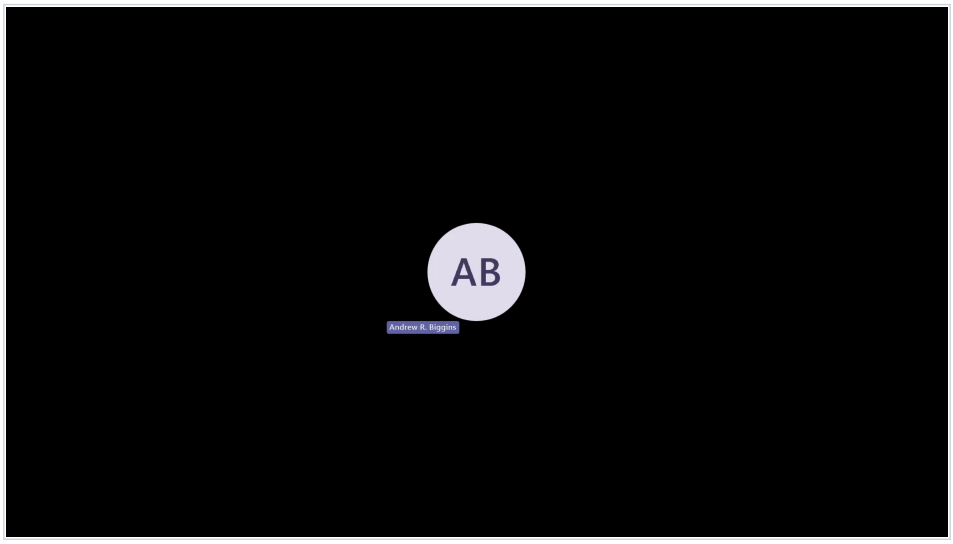


Figure 1 [00:00:05] — Teams recording › (camera off): Recording starts: “this video is how we update our survey feedback compliance”

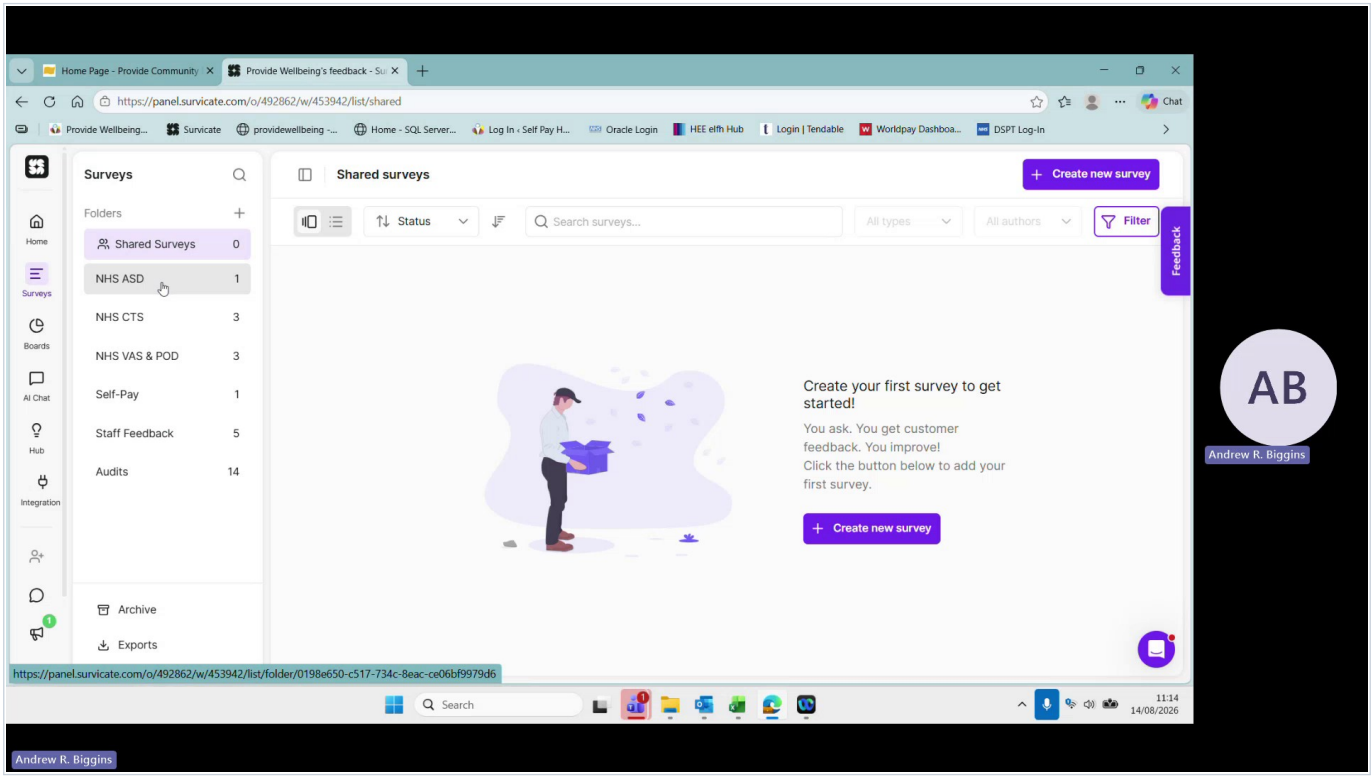


Figure 2 [00:00:48] — Survicate (web) › Shared surveys: Survicate survey platform (panel.survicate.com), opened from the Provide intranet bookmarks: folders NHS ASD (1), NHS CTS (3), NHS VAS & POD (3), Self-Pay (1), Staff Feedback (5), Audits (14)

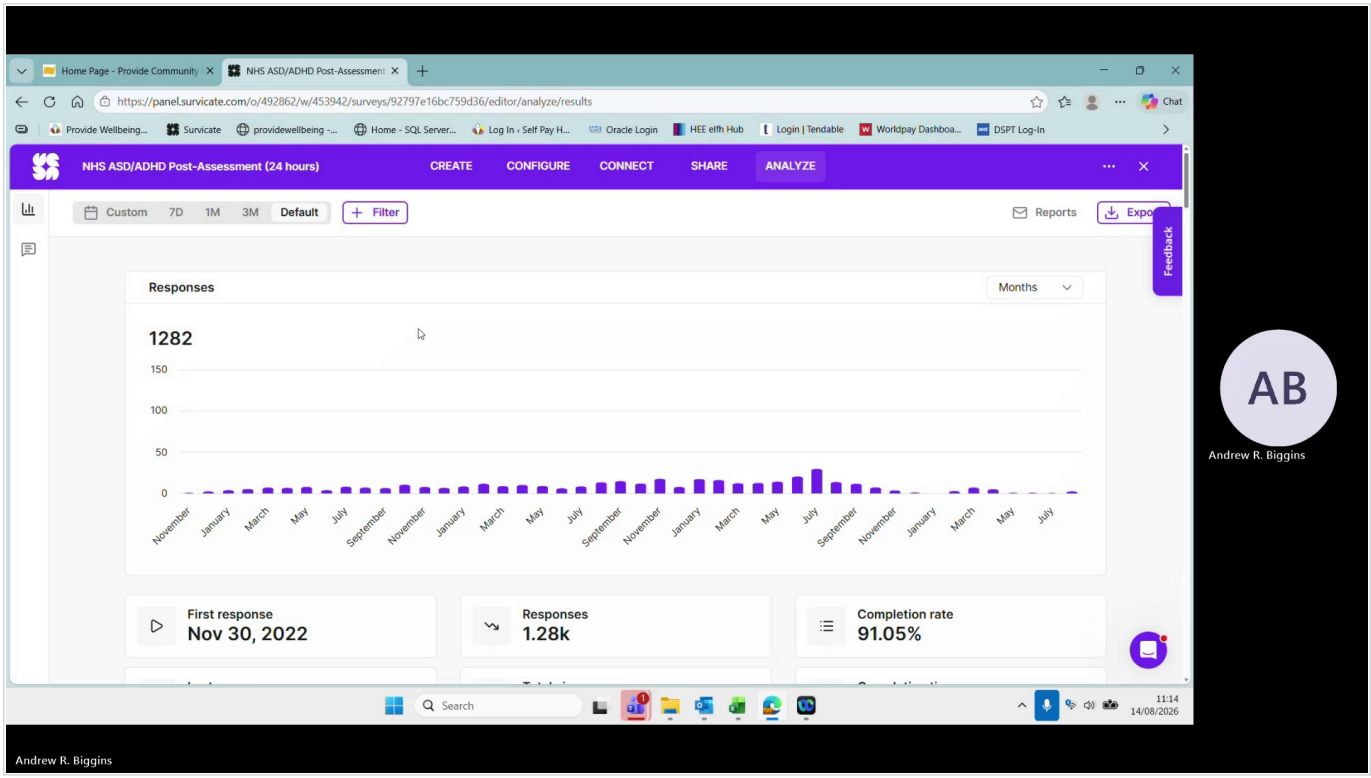


Figure 3 [00:01:15] — Survicate (web) › NHS ASD/ADHD Post-Assessment (24 hours): Survey results: 1,282 responses since 30 Nov 2022, 91.05% completion rate, monthly response chart; Export button top right

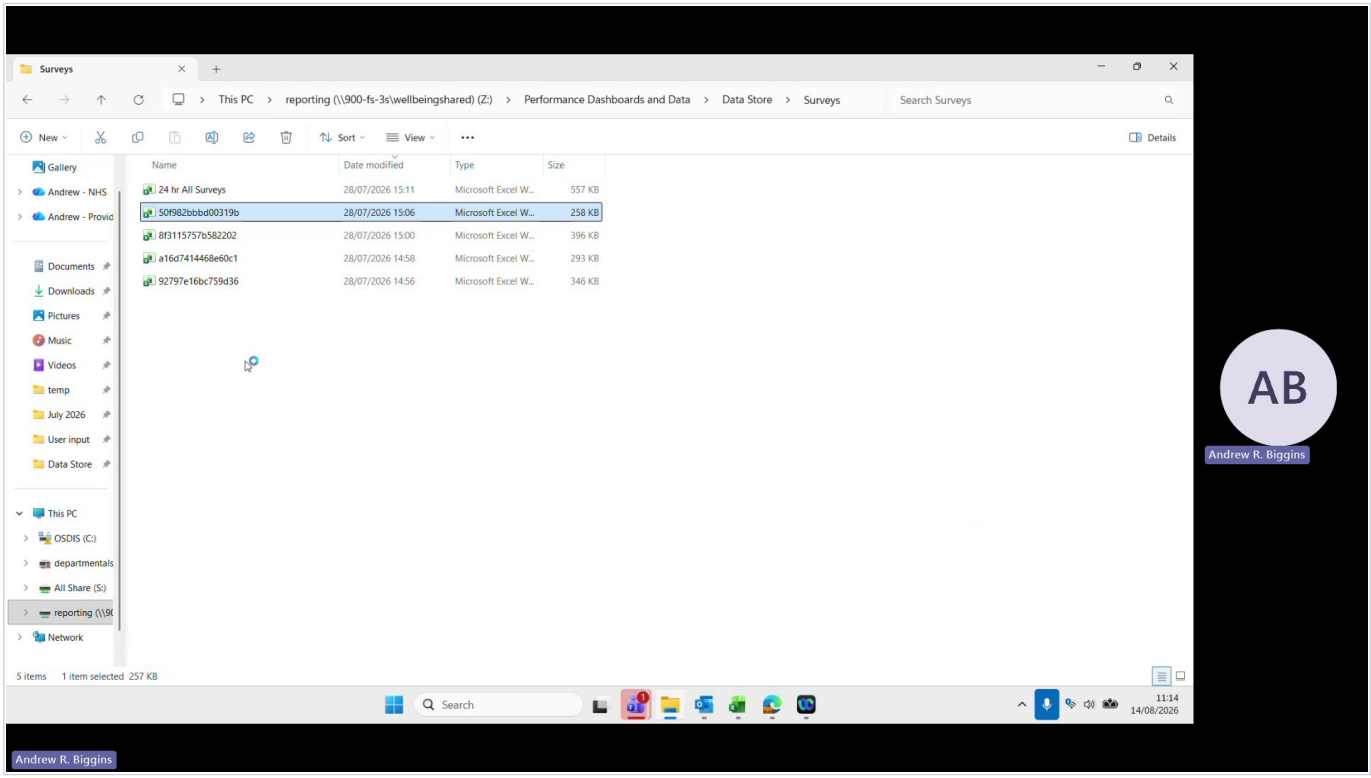


Figure 4 [00:01:24] — File Explorer › Z:\Performance Dashboards and Data\Data Store\Surveys: The Surveys folder: 24 hr All Surveys plus four exports with hash-like names (50f982bbbd00319b, 8f3115757b582202, a16d7414468e60c1, 92797e16bc759d36), all saved 28/07/2026

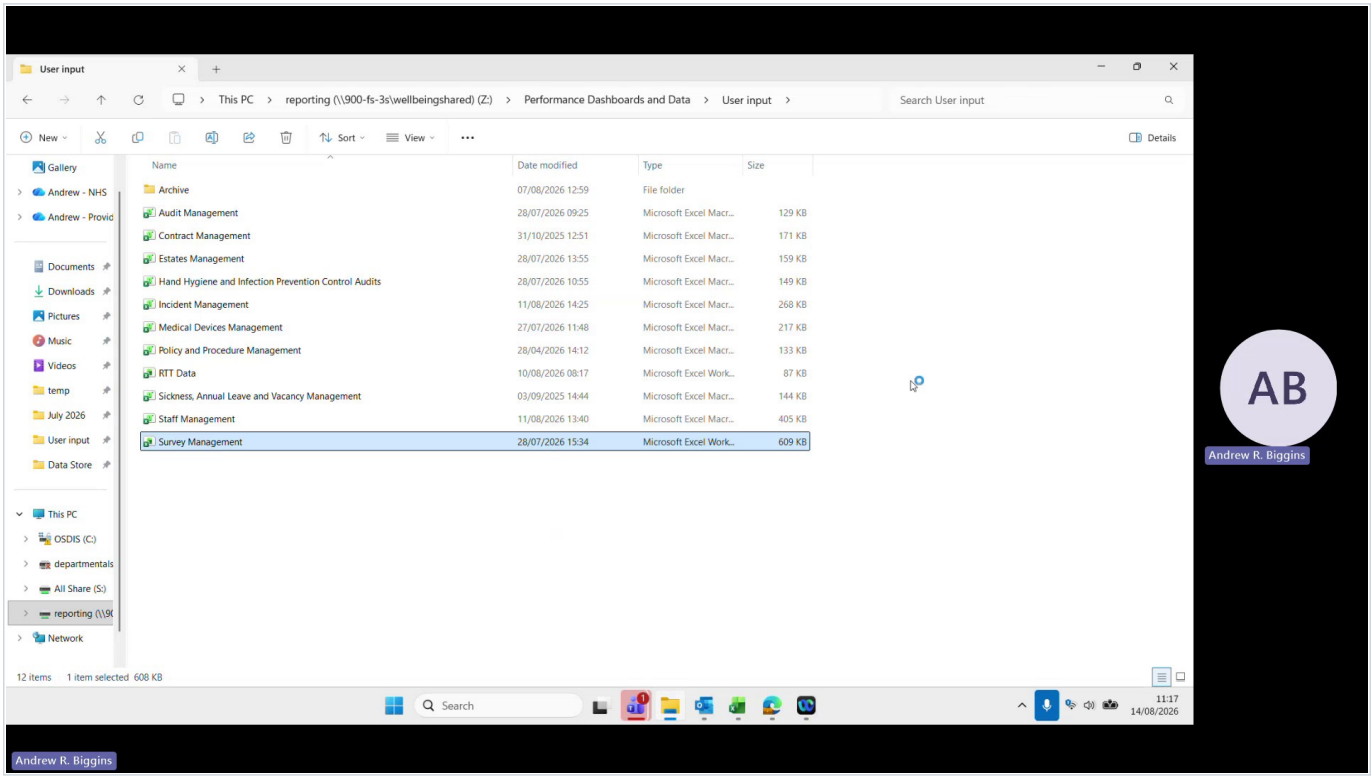


Figure 13 [00:04:27] — File Explorer › Z:\Performance Dashboards and Data\User input: User input folder: Survey Management.xlsm (second tool) beside Audit, Contract, Estates, Hand Hygiene/IPC, Incident, Medical Devices, Policy and Procedure, RTT Data, Sickness/Annual Leave/Vacancy, Staff Management

2. Exhaustive Column-Level Schema & Data Lineage Catalogue

2.1 Raw Survicate export (“Survey report”)

Col	Header	Type	Origin	Lineage, rule and evidence
A	Date & Time	Text datetime	RAW	e.g. 2022-08-03 1… (ISO text)
B–C	Response ID, Respondent	Text	RAW	Survicate identifiers
D–E	Device, Platform	Text	RAW	Desktop/Mobile; Win32, iPhone, Linux…
F	Page	URL	RAW	survey link; service name embedded (Ear Care, ASD and ADHD, No Scalpel Vasectomy…)
G	Q#1: What Provide Wellbeing service did you use?	Text	RAW	Self Pay survey only; mapped through Self Pay Types
H	Q#2: How … satisfied (1–5)	Number	RAW	Overall satisfaction
I	Q#3: How likely to recommend (1–5)	Number	RAW	Friends and Family
J	Q#4: free text	Text	RAW · **se nsitive**	Patient comment (blurred)
K–N	Q#5/Q#6 Action, “you for cor…”, “amp;”	Text	RAW	Action fields; a stray HTML-entity column “amp;” survives into the combined table

Row 2 is empty and “some of them clearly don’t… they click on the survey which registers a response but don’t fill it in”; these are removed by hand (00:01:18–00:01:49).

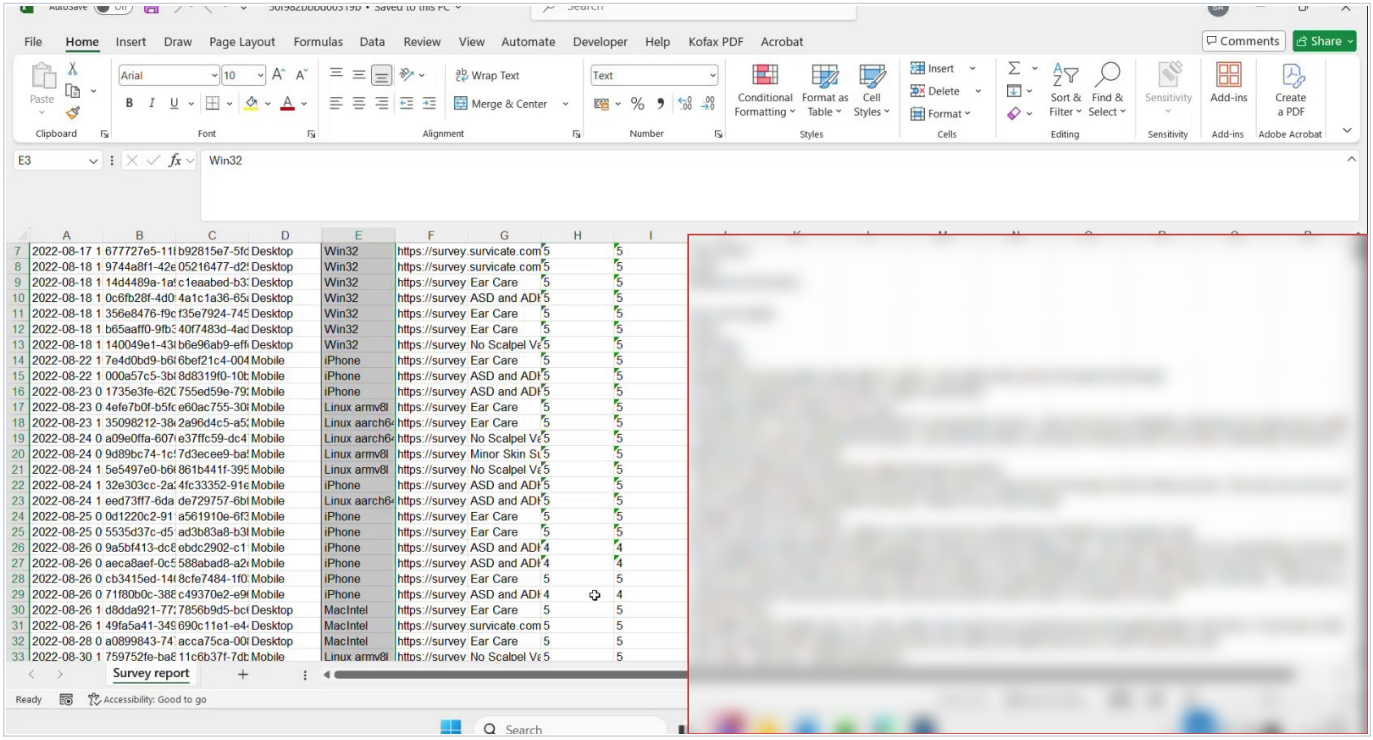


Figure 5 [00:01:36] — 50f982bbbd00319b.xlsx › Survey report: Raw Survicate export: Date & Time, Response ID, Respondent, Device, Platform, Page, Q#1 (overall satisfaction), Q#2 (friends and family), Q#3, free-text Q#4 (blurred), action columns; row 2 is empty

2.2 24 hr All Surveys (combined table, both tools)

Col	Header	Type	Origin	Lineage, rule and evidence
A	Date & Time (UTC)	Datetime	PQ	Parsed from export
B	Overall Satisfaction	1–5	PQ	From Q#2
C	Feedback	Text	PQ · **sens itive**	Comment (blurred)
D	Friends and Family	1–5	PQ	From Q#3

Col	Header	Type	Origin	Lineage, rule and evidence
E	amp;	Text	PQ	Artefact column
F	Service Level 2	Code	PQ	CTS, VAS, ND, or the Self Pay Types code
G	Service Level 1	Text	PQ	NHS / Self Pay

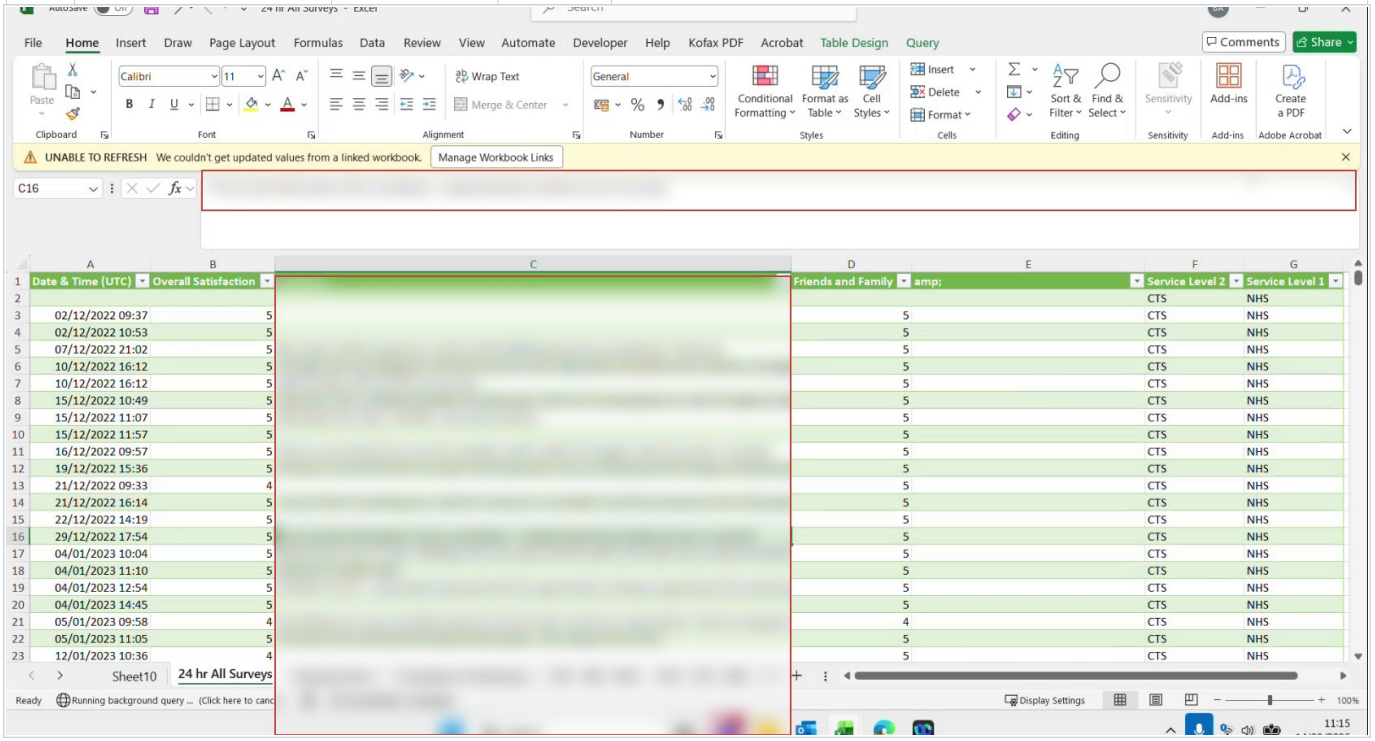


Figure 7 [00:02:30] — 24 hr All Surveys.xlsx › 24 hr All Surveys: 24 hr All Surveys table after refresh: Date & Time, Overall Satisfaction (1–5), Feedback (blurred), Friends and Family (1–5), “amp;” column, Service Level 2 (CTS, VAS, ND …), Service Level 1 (NHS / Self Pay). Status: Running background query; UNABLE TO REFRESH linked-workbook banner

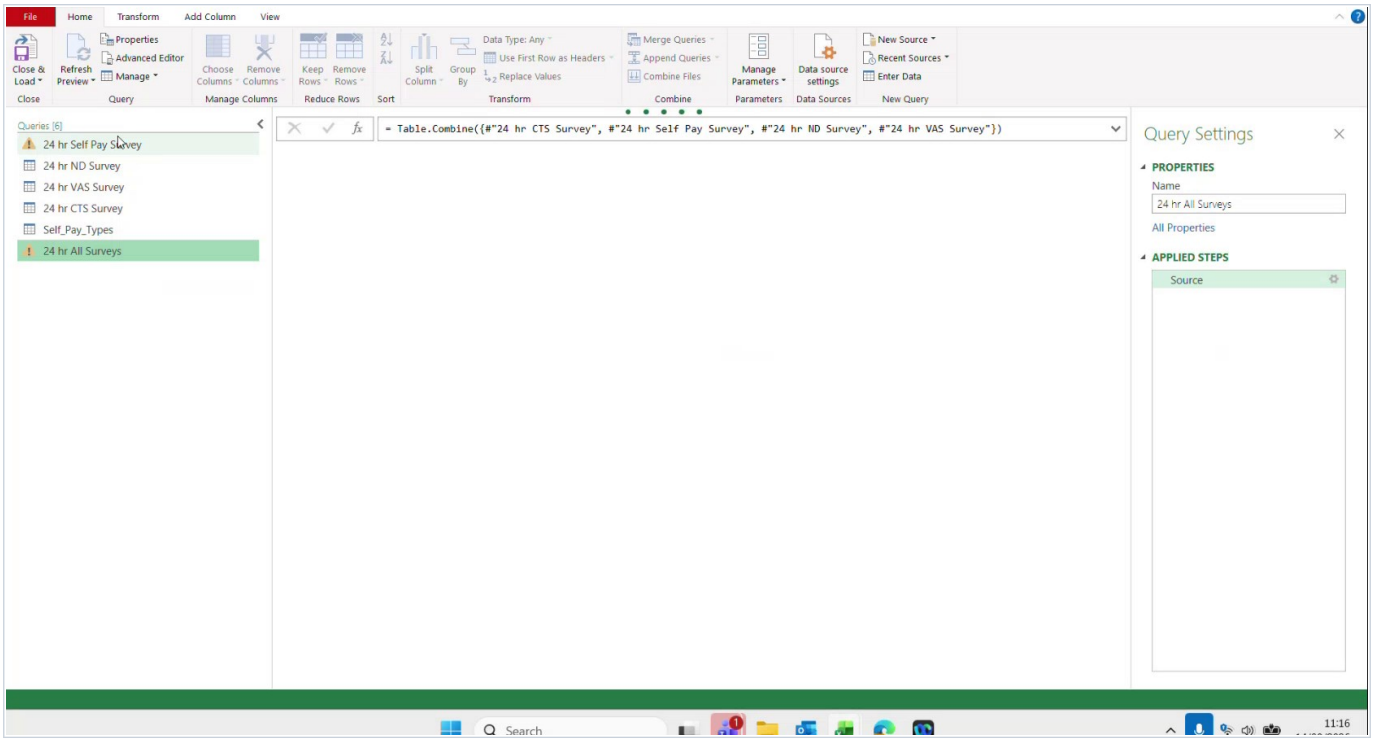


Figure 9 [00:03:18] — 24 hr All Surveys.xlsx › Power Query editor: Queries: 24 hr Self Pay Survey (warning icon), 24 hr ND/VAS/CTS Survey, Self_Pay_Types, 24 hr All Surveys = Table.Combine({CTS, Self Pay, ND, VAS}) (error icon) — “having trouble finding something with that one”

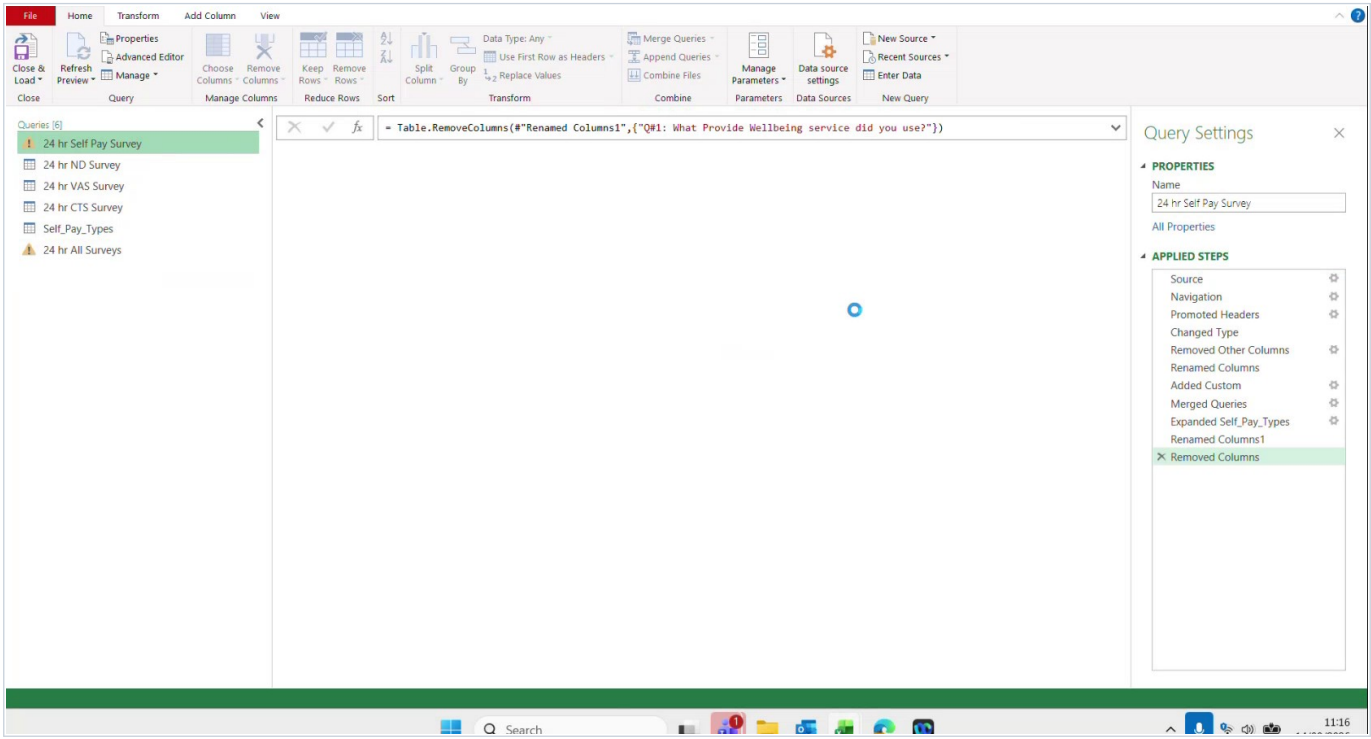


Figure 10 [00:03:22] — 24 hr All Surveys.xlsx › Power Query editor: 24 hr Self Pay Survey query (warning icon) with 11 applied steps (Source → Navigation → Promoted Headers → Changed Type → Removed Other Columns → Renamed Columns → Added Custom → Merged Queries → Expanded Self_Pay_Types → Renamed Columns1 → Removed Columns); last step =Table.RemoveColumns(#"Renamed Columns1",{"Q#1: What Provide Wellbeing service did you use?"})

2.3 Requirements and Compliance Timeframes

Both tools carry six compliance types for the single subject Provide Wellbeing: All, NHS, Self Pay, VAS, CTS, ND; Required TRUE, Core, Active TRUE. Timeframes: Every x months 0.5, People › Feedback, Unique Lookup `=CONCAT(Group 1,"-",Group 2,"-",Group 3)` → “People-Feedback-” (trailing dash), Missing TRUE. Type names differ between the tools (“SPC Flagged - All” vs “Feedback - SPC Flagged - All”).

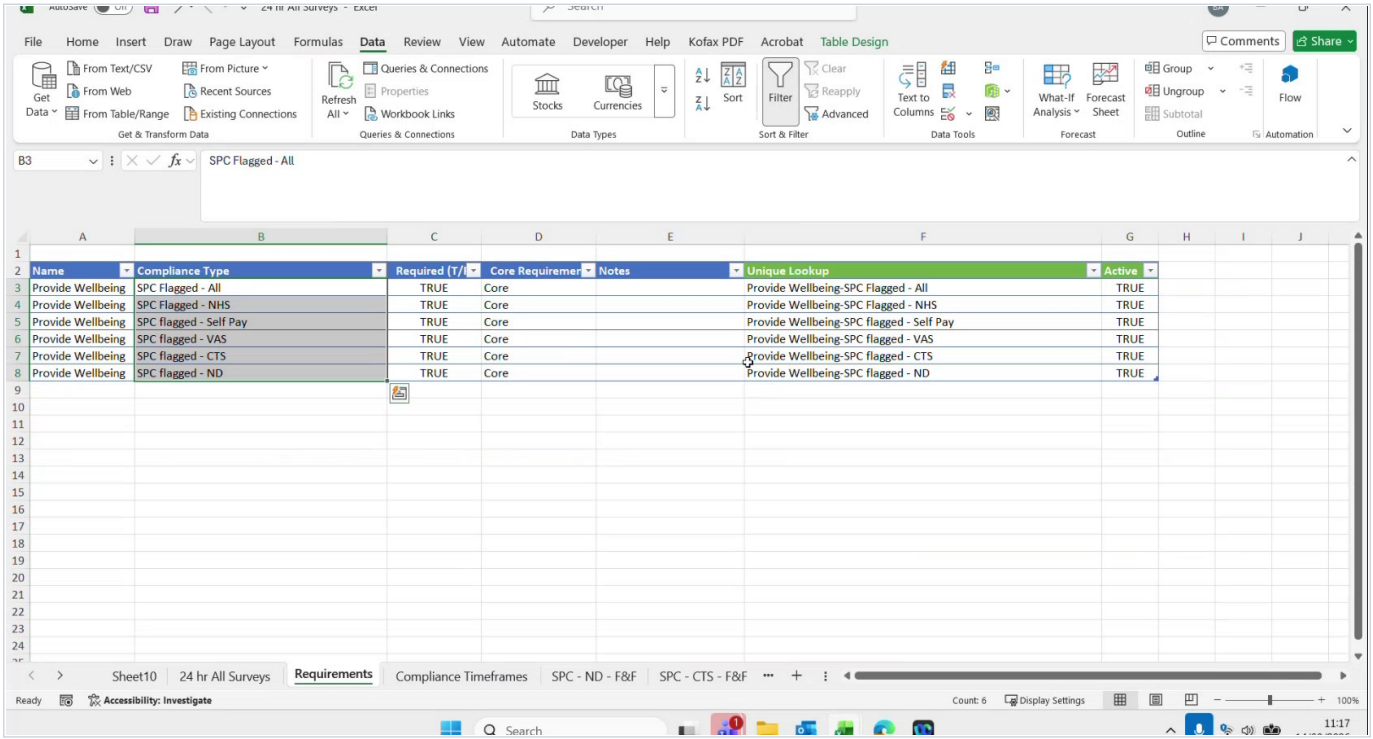


Figure 11 [00:04:05] — 24 hr All Surveys.xlsx › Requirements: Requirements in the first tool: Provide Wellbeing × SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND; Required TRUE, Core, Unique Lookup, Active TRUE

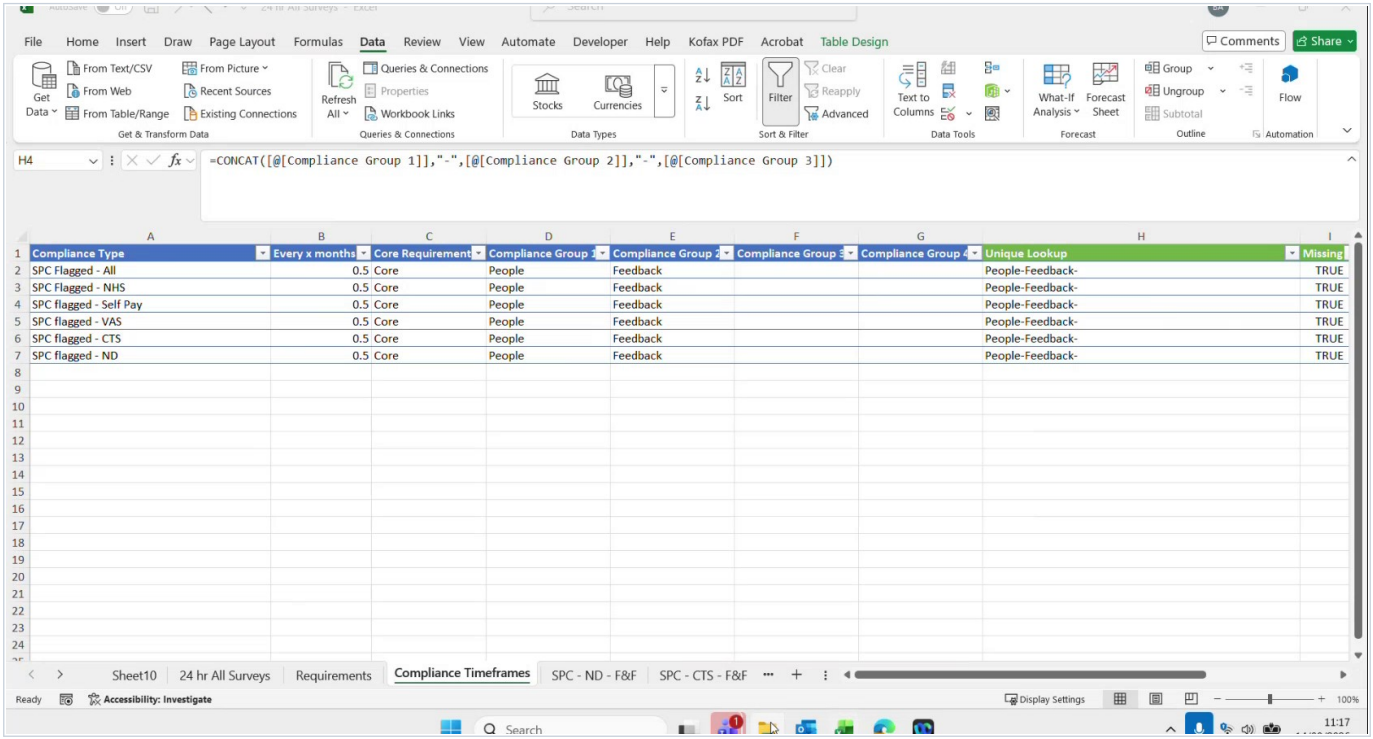


Figure 12 [00:04:16] — 24 hr All Surveys.xlsx › Compliance Timeframes: Compliance Timeframes in the first tool: H4 Unique Lookup =CONCAT(Group 1,"-",Group 2,"-",Group 3) → “People-Feedback-”; Every x months 0.5; Missing TRUE

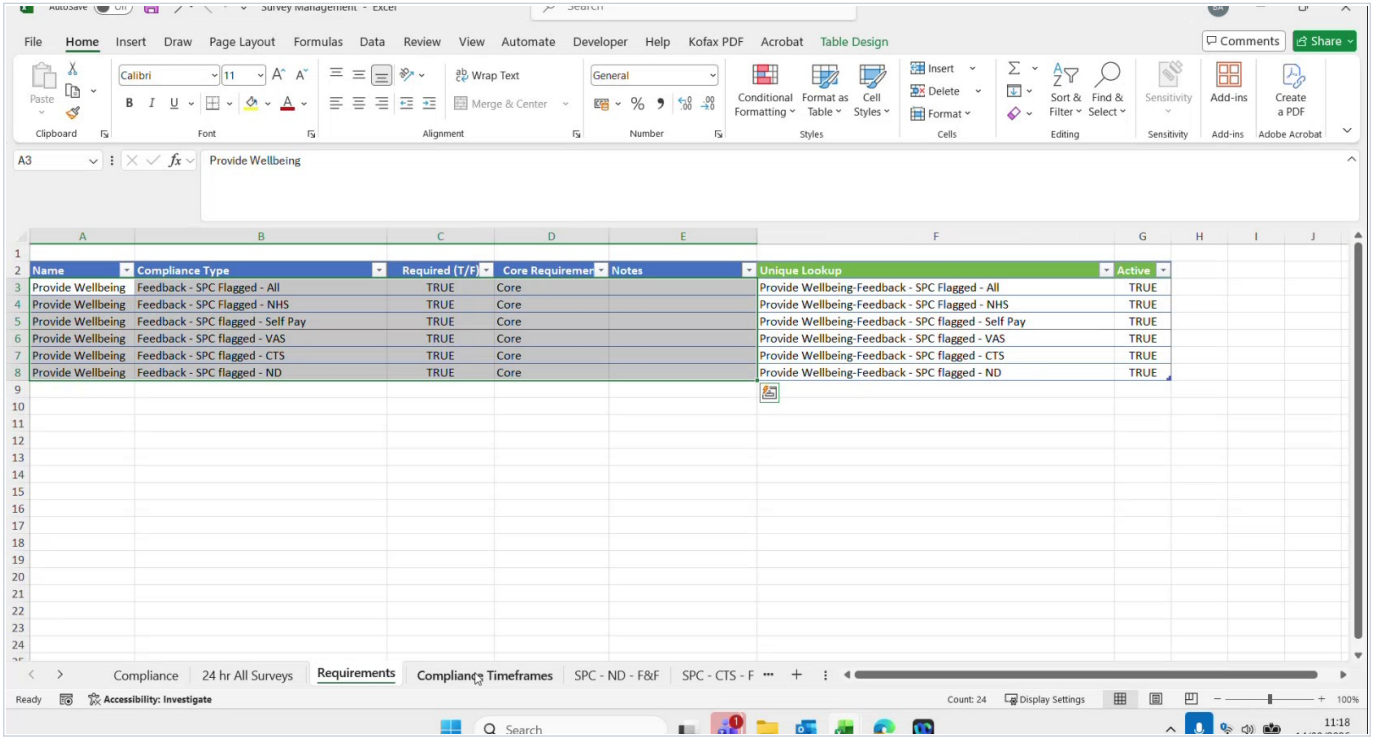


Figure 14 [00:04:40] — Survey Management.xlsm › Requirements: Survey Management has its own Requirements (and Compliance Timeframes): Compliance Type “Feedback - SPC Flagged - All” etc., prefixed “Feedback -” unlike the first tool’s “SPC Flagged - All”; Required TRUE, Core, Active TRUE

2.4 SPC sheets (F&F and CSAT, per service)

Column(s)	Content	Rule
B	Month (1st of month)	From 01/09/2024 (first tool) / earlier (second tool) to last month
C	Monthly average score	Average of Friends and Family (F&F sheets) or Overall Satisfaction (CSAT sheets) for the service
F, G, H	Mean, LCL, UCL (repeated per row)	AC3 mean, AC4 standard deviation, `AC5 LCL =AC3-(AC4*3)`, AC6 UCL = mean + 3σ
I, J	Moving range, cumulative moving average	For the σ estimate [Inferred]

Column(s)	Content	Rule
K–Z	Count > 3, Count < 3, Under/Over Mean, 7 points </> mean, Under LCL, Over UCL, Trend Up/Down, 7-point trends, 2 out of 3 low/high	Each 1 when the rule fires; e.g. N19 =IF(AND(N18=1,M19=1),IF(SUM(M19:M25)>=7,1,0),0)

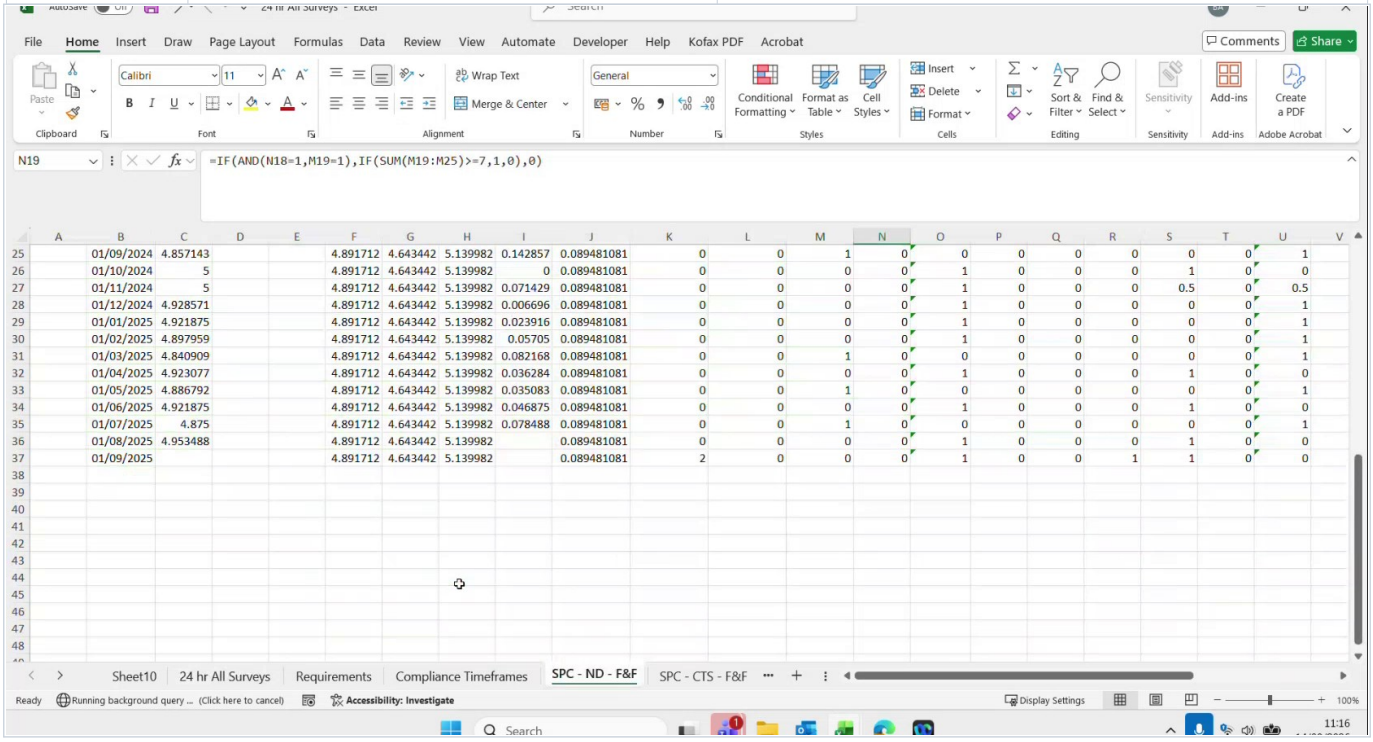


Figure 8 [00:03:01] — 24 hr All Surveys.xlsx › SPC - ND - F&F: SPC sheet in the first tool: monthly averages from 01/09/2024 and the rule columns; N19 =IF(AND(N18=1,M19=1),IF(SUM(M19:M25)>=7,1,0),0) (7 points below the mean)

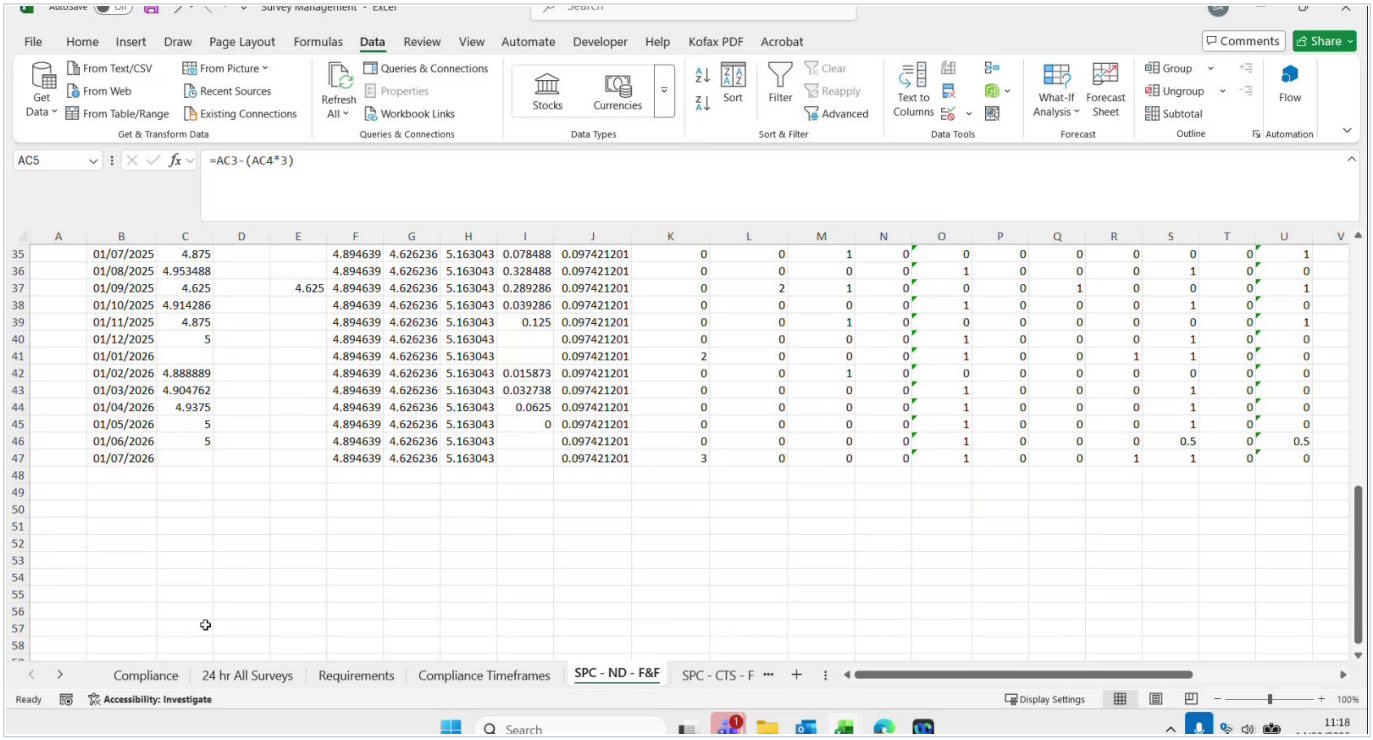


Figure 15 [00:05:21] — Survey Management.xlsm › SPC - ND - F&F: SPC - ND - F&F: monthly mean score per month (to 01/07/2026), mean / LCL / UCL, cumulative moving average and the rule flags; AC5 LCL =AC3-(AC4*3)

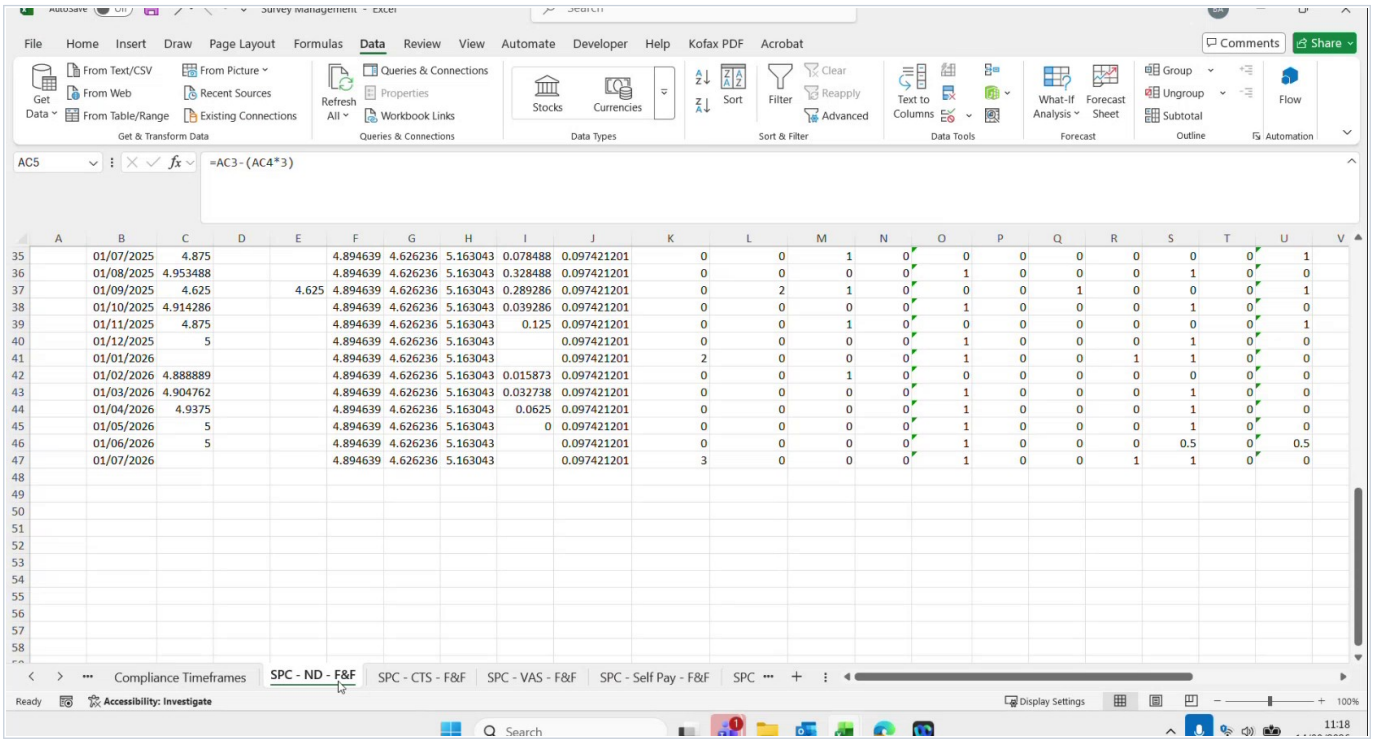


Figure 16 [00:05:25] — Survey Management.xlsm › SPC - Self Pay - F&F: SPC rule columns: Count > 3, Count < 3, Under Mean, 7 points < mean, Over Mean, 7 points > mean, Under LCL, Over UCL, Trend Up, 7 points Trend Up, Trend Down, 7 points Trend Down, 2 out of 3 low, 2 out of 3 high; Mean 4.81, StdDev 0.105, LCL 4.50, UCL 5.12

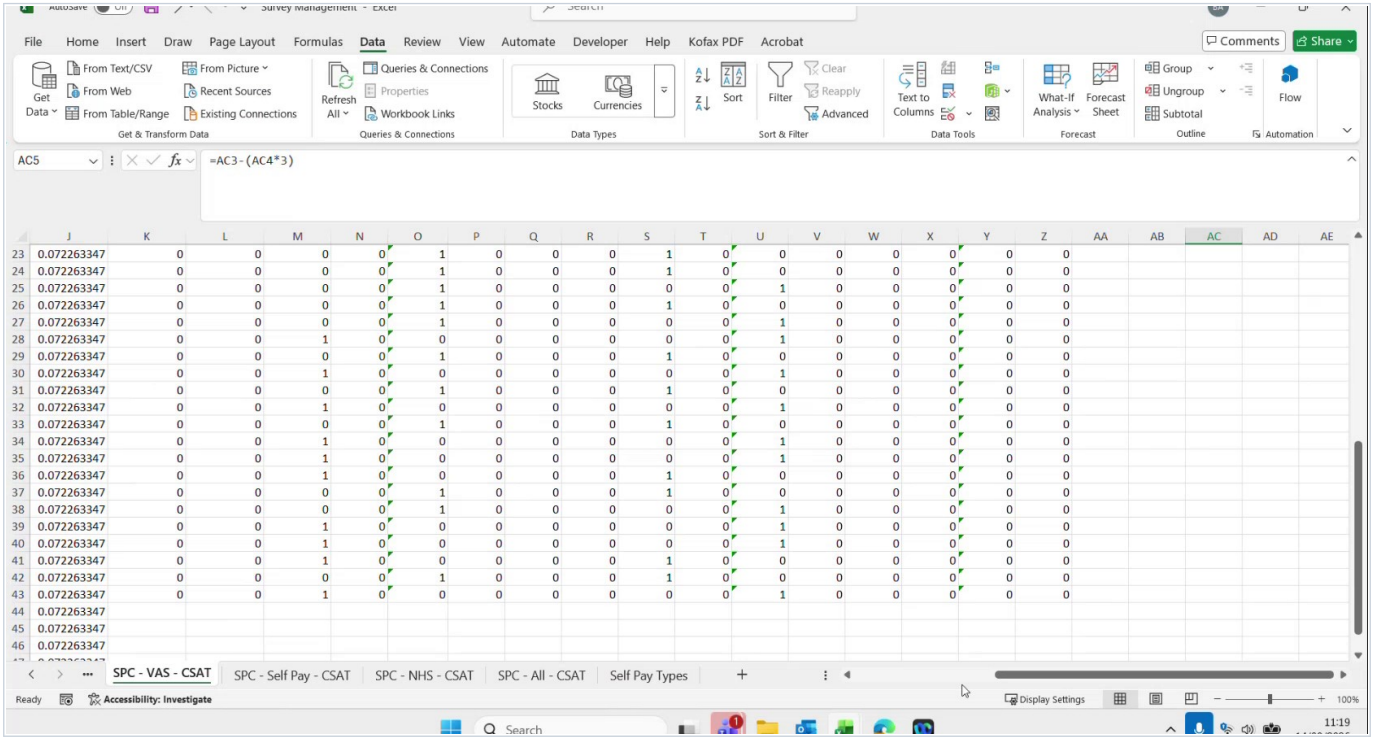


Figure 17 [00:05:48] — Survey Management.xlsm › SPC - VAS - CSAT: CSAT sheets (VAS, Self Pay, NHS, All) follow the same SPC layout; tab list ends with Self Pay Types

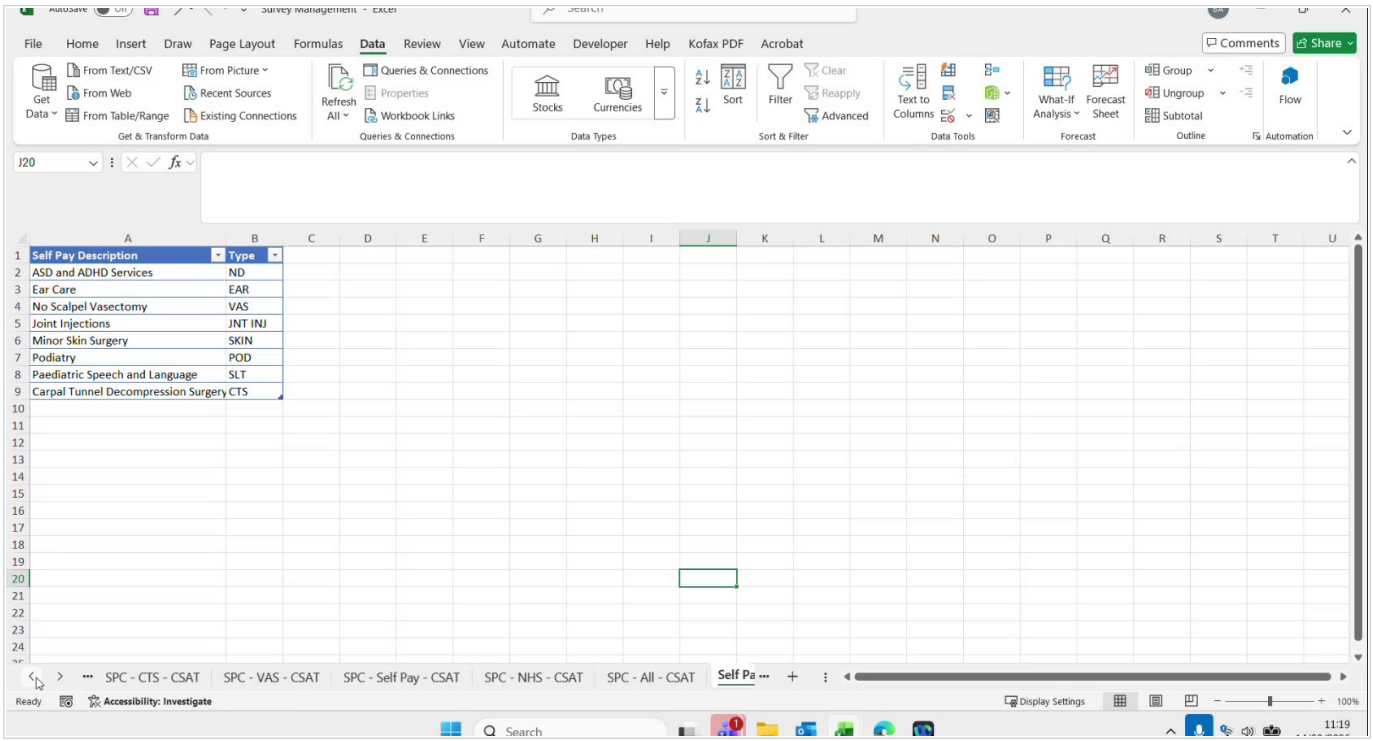


Figure 18 [00:06:01] — Survey Management.xlsm › Self Pay Types: Self Pay Types: service description → code (ASD and ADHD → ND, Ear Care → EAR, No Scalpel Vasectomy → VAS, Joint Injections → JNT INJ, Minor Skin Surgery → SKIN, Podiatry → POD, Paediatric Speech and Language → SLT, Carpal Tunnel → CTS)

2.5 Survey Management › Compliance (output rows)

Col	Header	Type	Origin	Lineage, rule and evidence
A	Name	Text	MAN	Provide Wellbeing
B	Compliance Type	Text	MAN	Feedback - SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND
C	Completion Date	Date	MAN/CALC	14/08/2026 (update day)
D	Completion Status	Text	MAN	Completed
E	RAG Rating	Number	MAN	10
F	Notes	Text	CALC	Lists flagged measures with month, else “Satisfaction and F&F within SPC criteria”. Figure 19 [00:06:07]
G–H	Managed By, Due Date	Text, Date	MAN	Provide Wellbeing; 24/08/2026
I–M	Compliance Group 1–4, Weighting	Text	MAN	People › Feedback; 1
N–O	SPC Count CSAT, SPC Count F&F	Integer	CALC	Rule hits for last month
P–Q	Avg CSAT, Avg F&F	Number	CALC	`=XLOOKUP(EDATE(DATE(YEAR(TODAY()),MONTH(TODAY()),1),-1),'SPC - Self Pay - CSAT'!B:B,…C:C,"")`. Figure 20 [00:06:14]
R…	(cross-tool)	Number	CALC	Reads Incident Management.xlsm Estates by full path. Figure 21 [00:06:17]

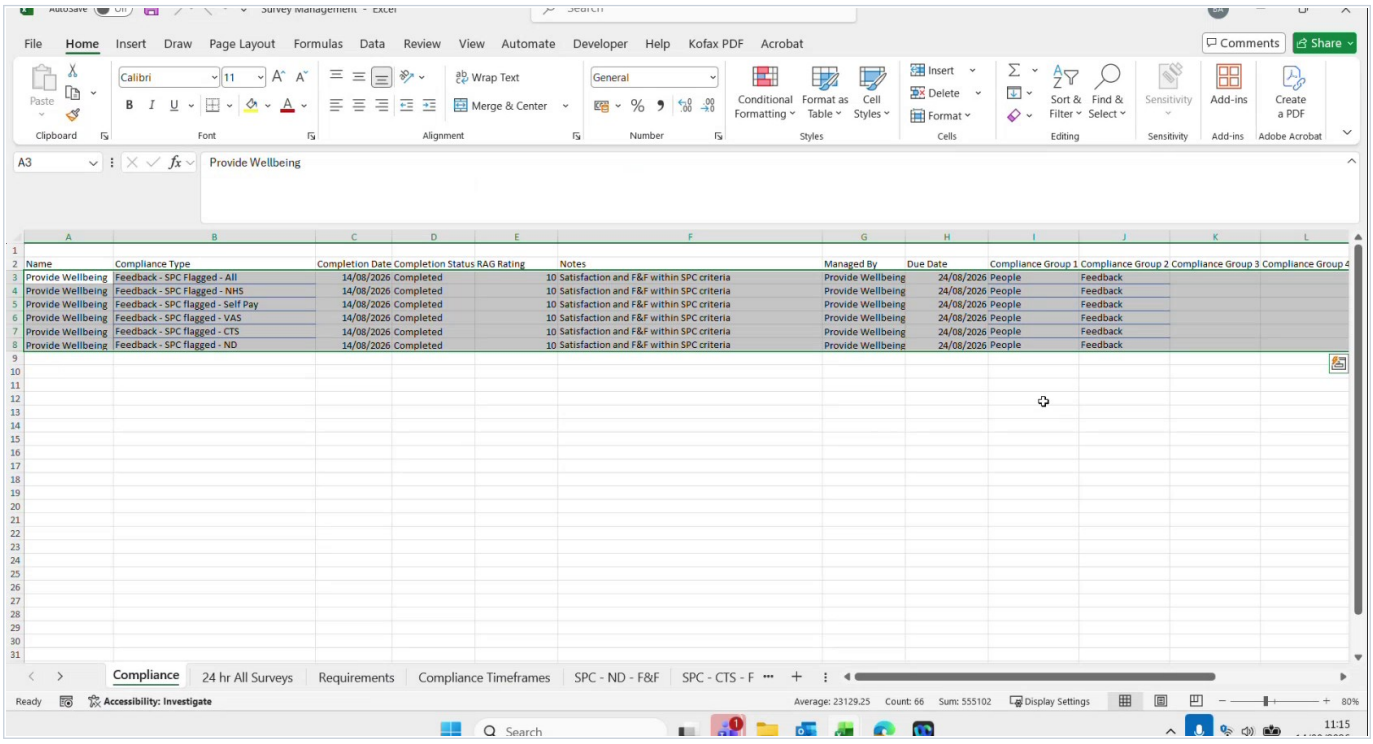


Figure 6 [00:02:21] — Survey Management.xlsm › Compliance: Survey Management.xlsm opens on Compliance: six rows (Feedback - SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND), completed 14/08/2026, Issue 10, “Satisfaction and F&F within SPC criteria”, due 24/08/2026, People › Feedback

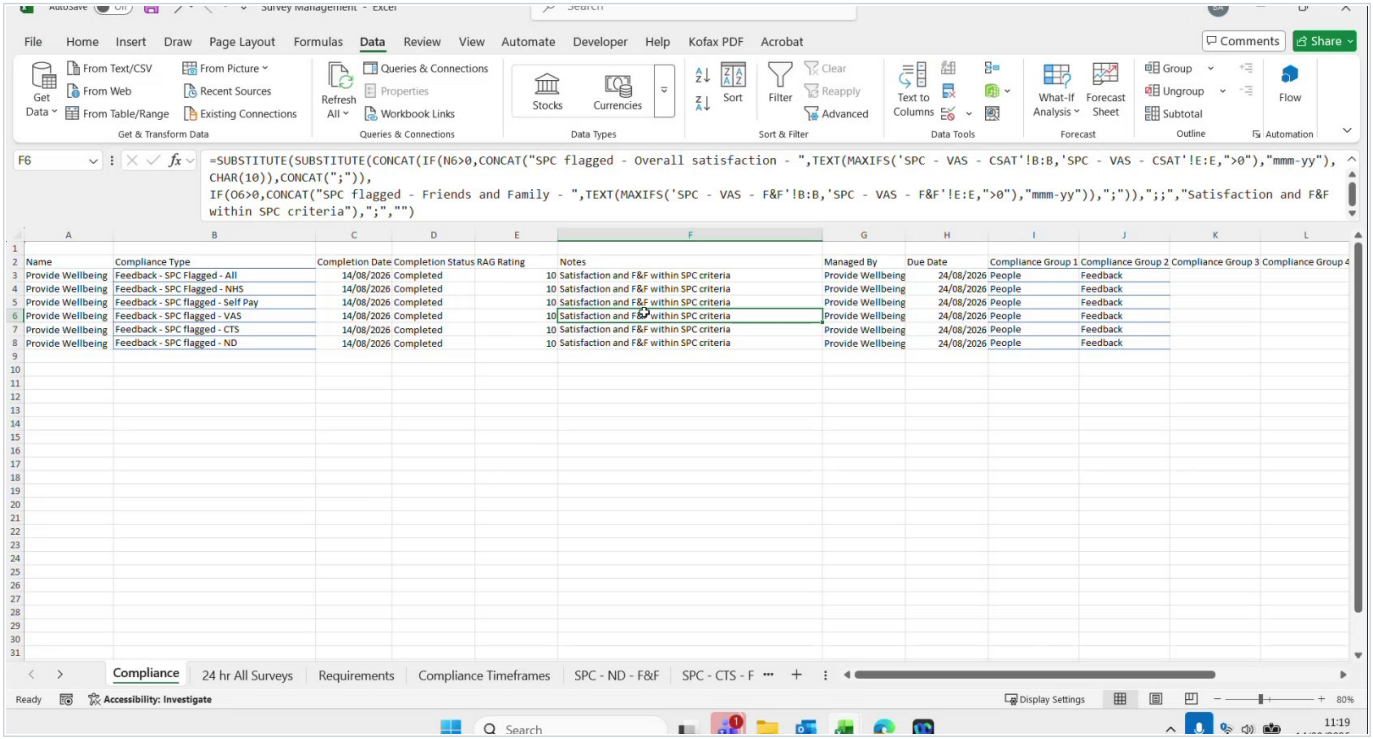


Figure 19 [00:06:07] — Survey Management.xlsm › Compliance: Formula bar, F6 Notes: SUBSTITUTE(…CONCAT(IF(N6>0,"SPC flagged - Overall satisfaction - "&TEXT(MAXIFS('SPC - VAS - CSAT'!B:B,'SPC - VAS - CSAT'!E:E,">0"),"mmm-yy")…),IF(O6>0,"SPC flagged - Friends and Family - …")…),"","Satisfaction and F&F within SPC criteria")

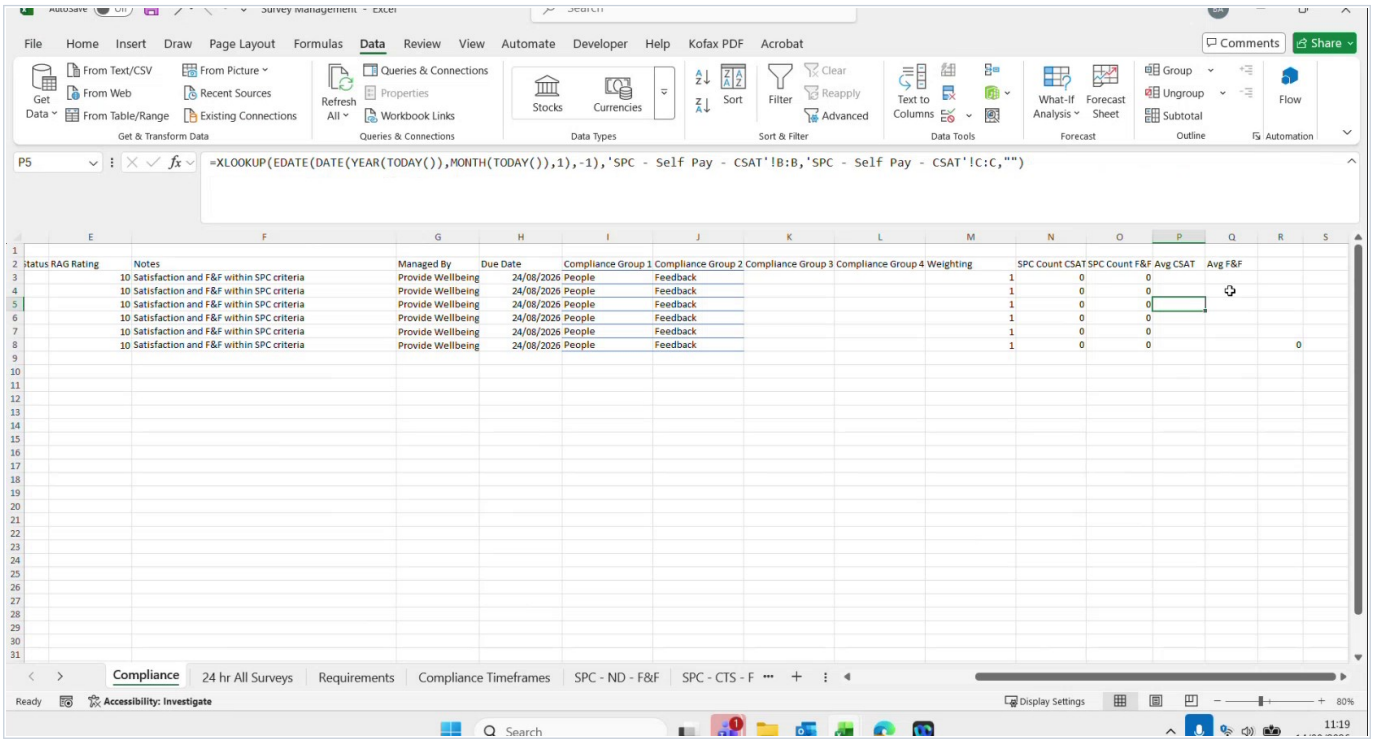


Figure 20 [00:06:14] — Survey Management.xlsm › Compliance: Formula bar, P5: =XLOOKUP(EDATE(DATE(YEAR(TODAY()),MONTH(TODAY()),1),-1),'SPC - Self Pay - CSAT'!B:B,'SPC - Self Pay - CSAT'!C:C,"") — last month's average; columns N:Q SPC Count CSAT, SPC Count F&F, Avg CSAT, Avg F&F

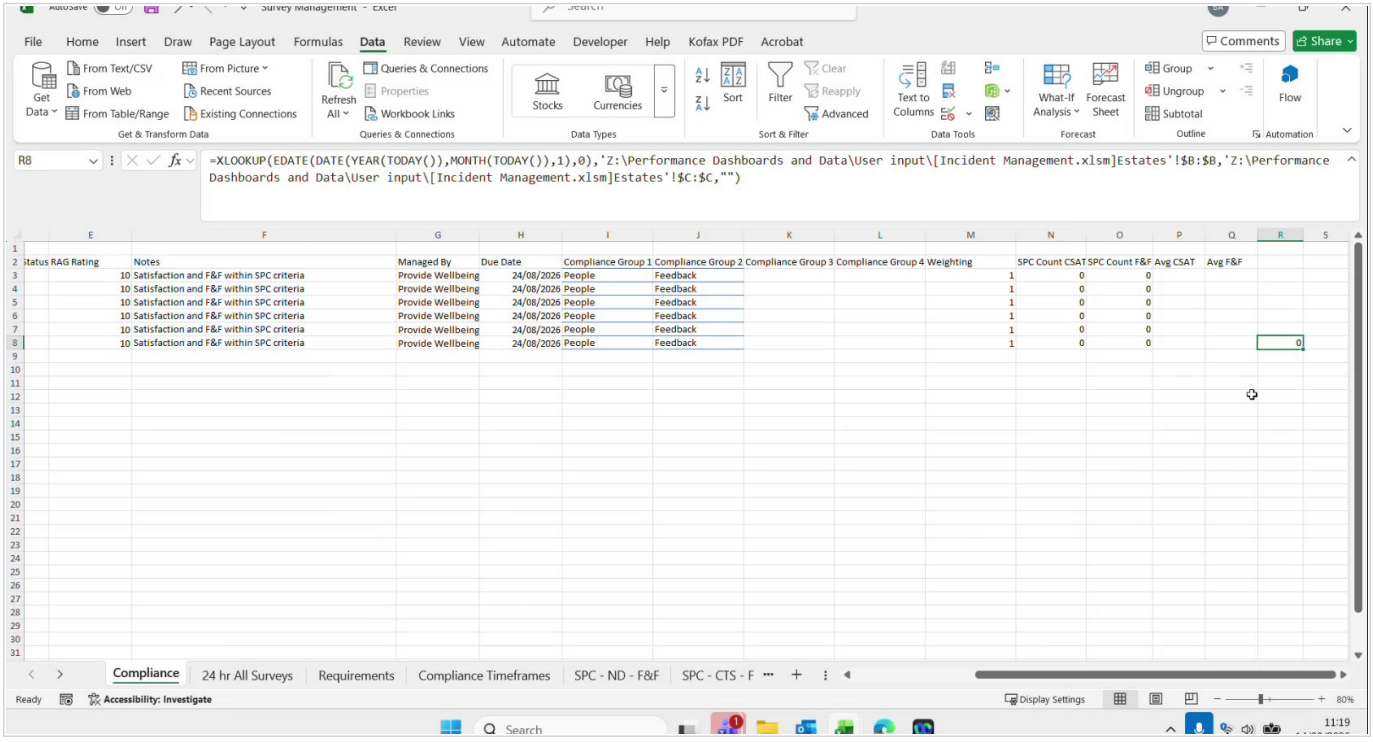


Figure 21 [00:06:17] — Survey Management.xlsm › Compliance: Formula bar, R8: =XLOOKUP(EDATE(DATE(YEAR(TODAY()),MONTH(TODAY()),1),0),'Z:\Performance Dashboards and Data\User input\[Incident Management.xlsm]Estates'!$B:$B, …) — a link into the Incident Management workbook's Estates tab

2.6 Compliance Data.xlsm

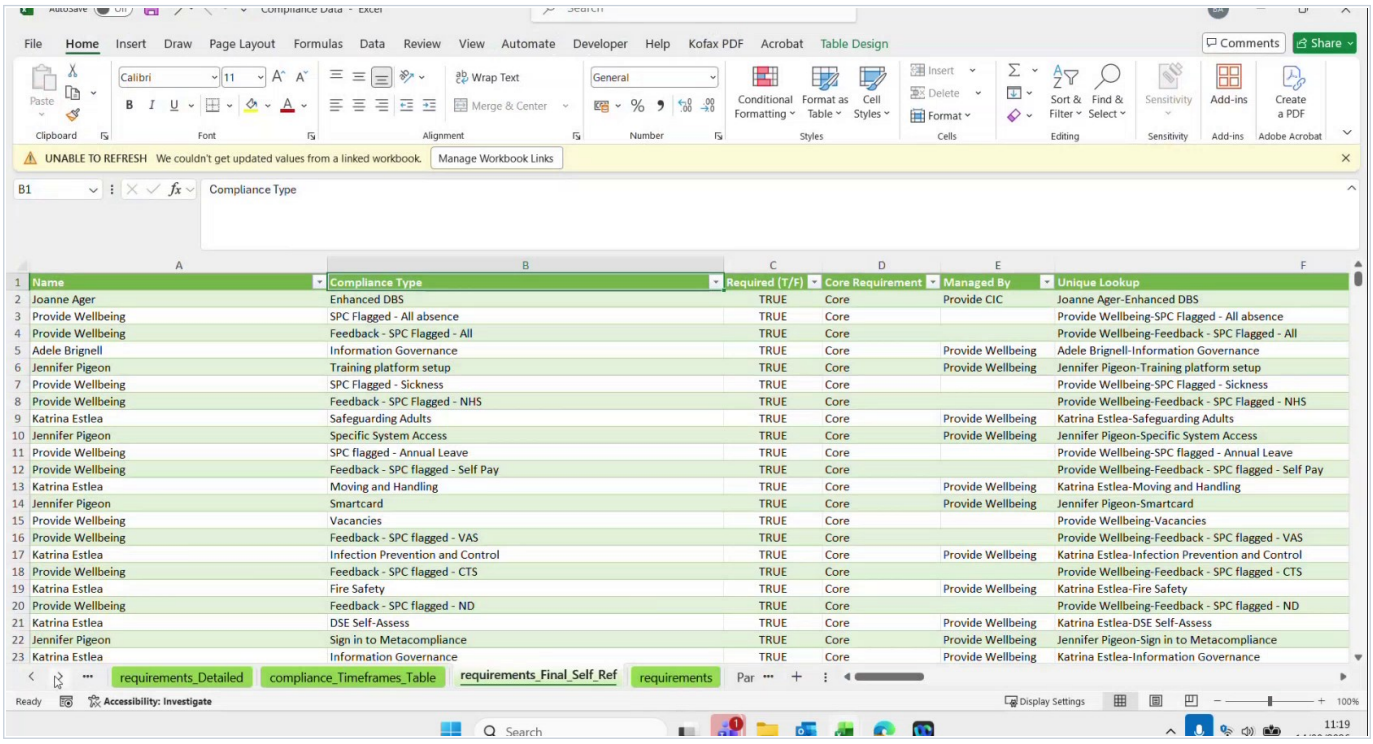


Figure 22 [00:06:36] — Compliance Data.xlsm › requirements_Final_Self_Ref: Data store requirements (UNABLE TO REFRESH banner): Feedback - SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND rows next to staff rows

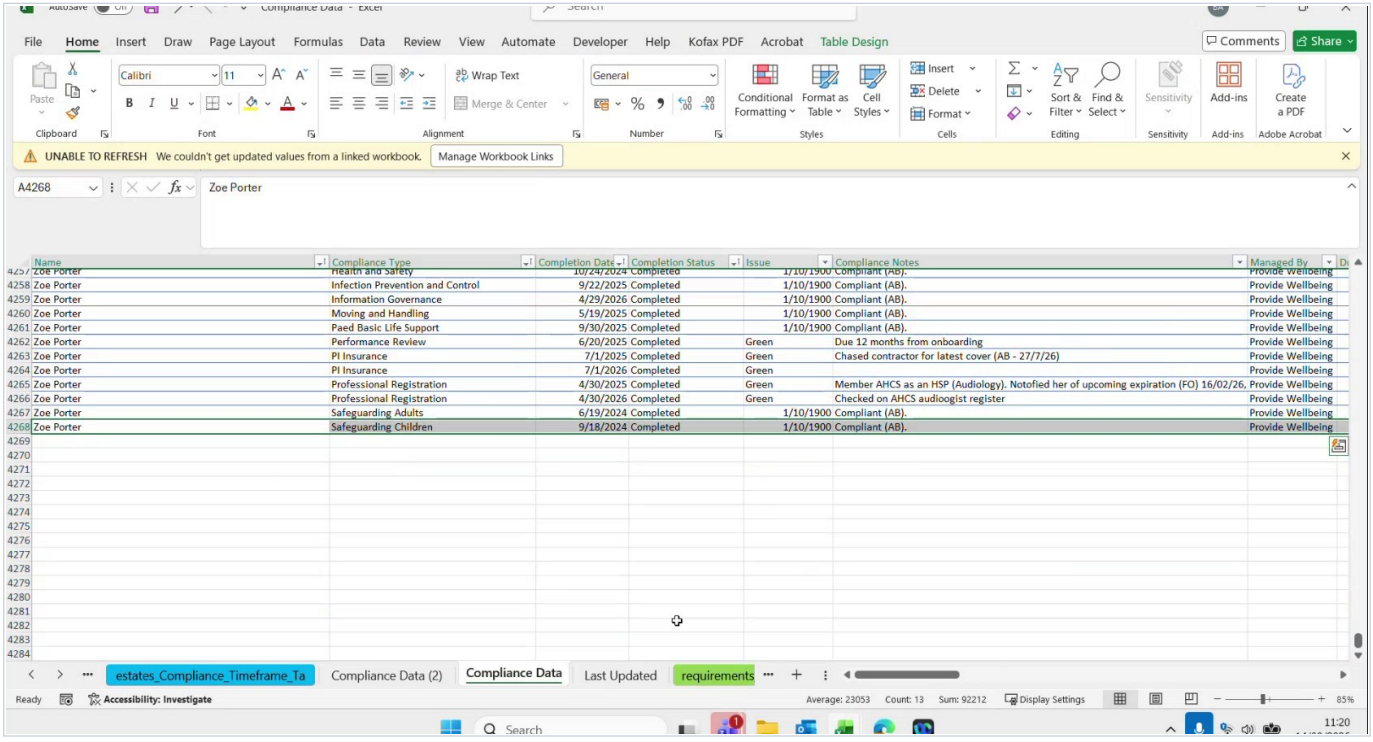


Figure 23 [00:06:42] — Compliance Data.xlsm › Compliance Data: Compliance Data last row 4268: where the six Compliance rows are pasted (paste not performed on screen)

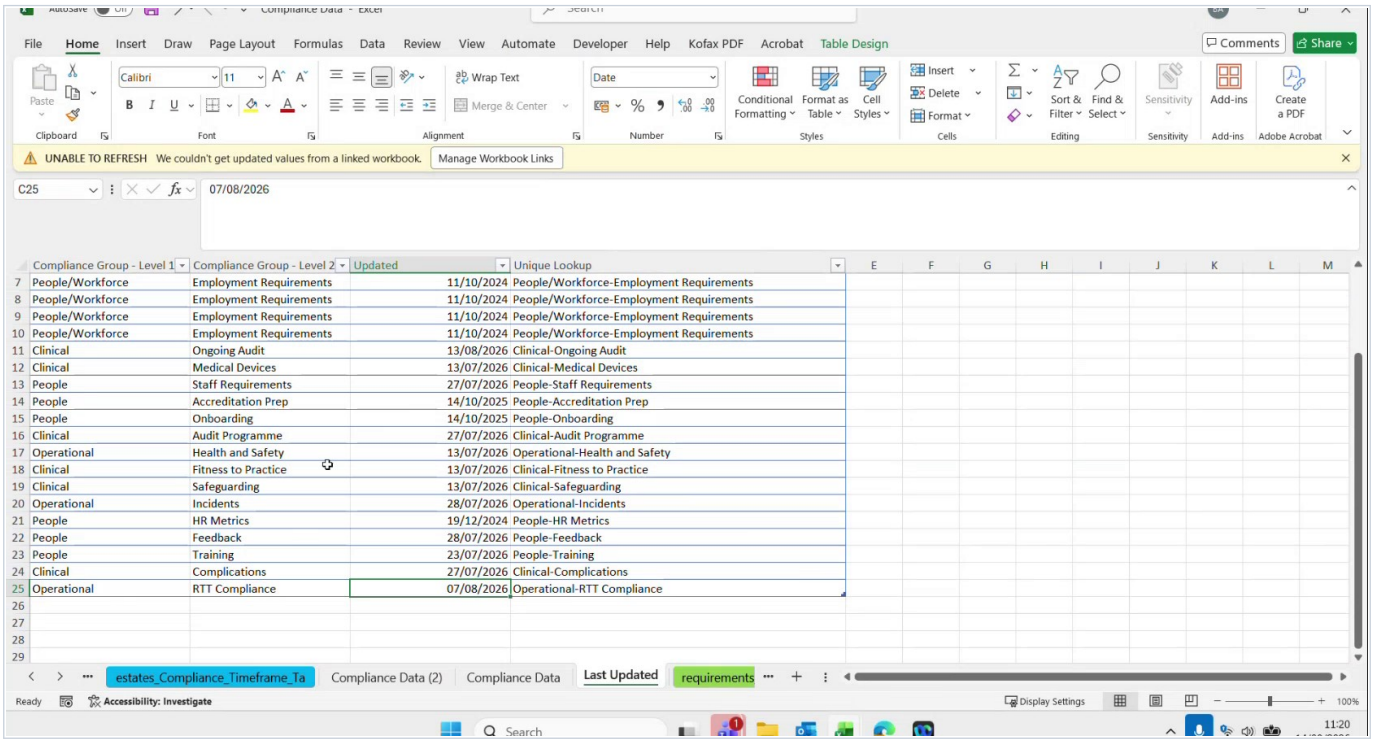


Figure 24 [00:06:46] — Compliance Data.xlsm › Last Updated: Last Updated: People-Feedback updated 28/07/2026

3. Chronological Step-by-Step Operator Workflow

Nine steps cover the 7:07 recording. The download, cleaning, month roll-forward and paste are described, not performed.

Step	Window	Action
1	00:00:00–00:00:23	Introduce survey feedback compliance
2	00:00:23–00:01:11	Open Survicate and explain the four surveys
3	00:01:11–00:01:57	Download and clean the exports (described)
4	00:01:57–00:03:50	Open the first tool and refresh
5	00:03:50–00:04:25	Requirements and Compliance Timeframes (first tool)
6	00:04:25–00:05:08	Second tool: Survey Management
7	00:05:08–00:06:03	SPC tables for F&F and CSAT
8	00:06:03–00:06:20	Compliance table
9	00:06:20–00:07:07	Paste into the data store, last updated, save, refresh twice (described)

Step 1 [00:00:00 – 00:00:23]: Introduce survey feedback compliance

Action performed	Camera off; starts screen share.
Source → target	—
State mutation	—

Shown above: Figure 1 [00:00:05] — Recording starts: “this video is how we update our survey feedback compliance”

Step 2 [00:00:23 – 00:01:11]: Open Survicate and explain the four surveys

Action performed	From the intranet bookmark opens Survicate; four 24-hour surveys; satisfaction and F&F out of five; “four and five is acceptable”.
Source → target	—
State mutation	—

Shown above: Figure 2 [00:00:48] — Survicate survey platform (panel.survicate.com), opened from the Provide intranet bookmarks: folders NHS ASD (…

Shown above: Figure 3 [00:01:15] — Survey results: 1,282 responses since 30 Nov 2022, 91.05% completion rate, monthly response chart; Export butt…

Step 3 [00:01:11 – 00:01:57]: Download and clean the exports (described)

Action performed	Shows last month’s exports in Data Store\Surveys and one raw file; spoilt responses removed by hand.
Source → target	Survicate → .xlsx exports → manual clean → Surveys folder
State mutation	Not performed live

Shown above: Figure 4 [00:01:24] — The Surveys folder: 24 hr All Surveys plus four exports with hash-like names (50f982bbbd00319b, 8f3115757b5822…

Shown above: Figure 5 [00:01:36] — Raw Survicate export: Date & Time, Response ID, Respondent, Device, Platform, Page, Q#1 (overall satisfaction)…

Step 4 [00:01:57 – 00:03:50]: Open the first tool and refresh

Action performed	Opens a workbook (Survey Management shown first, then 24 hr All Surveys); Data › Refresh All runs a background query; Power Query editor shows the Self Pay query warning and the combined query error.
Source → target	Exports → 24 hr All Surveys (Power Query)
State mutation	Table refreshed; one query failing

Shown above: Figure 6 [00:02:21] — Survey Management.xlsm opens on Compliance: six rows (Feedback - SPC Flagged - All / NHS / Self Pay / VAS / CT…

Shown above: Figure 7 [00:02:30] — 24 hr All Surveys table after refresh: Date & Time, Overall Satisfaction (1–5), Feedback (blurred), Friends an…

Shown above: Figure 8 [00:03:01] — SPC sheet in the first tool: monthly averages from 01/09/2024 and the rule columns; N19 =IF(AND(N18=1,M19=1),I…

Shown above: Figure 9 [00:03:18] — Queries: 24 hr Self Pay Survey (warning icon), 24 hr ND/VAS/CTS Survey, Self_Pay_Types, 24 hr All Surveys = Ta…

Shown above: Figure 10 [00:03:22] — 24 hr Self Pay Survey query (warning icon) with 11 applied steps (Source → Navigation → Promoted Headers → Cha…

Step 5 [00:03:50 – 00:04:25]: Requirements and Compliance Timeframes (first tool)

Action performed	“We have our usual requirements and compliance timeframes. This is a bit confusing.”
Source → target	—
State mutation	—

Shown above: Figure 11 [00:04:05] — Requirements in the first tool: Provide Wellbeing × SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND; Requi…

Shown above: Figure 12 [00:04:16] — Compliance Timeframes in the first tool: H4 Unique Lookup =CONCAT(Group 1,"-",Group 2,"-",Group 3) → "People-F…

Step 6 [00:04:25 – 00:05:08]: Second tool: Survey Management

Action performed	Opens Survey Management.xlsm from User input: "also has requirements and compliance timeframes... I'm not even sure we need to complete the first one."
Source → target	Exports → Survey Management (Power Query)
State mutation	—

Shown above: Figure 13 [00:04:27] — User input folder: Survey Management.xlsm (second tool) beside Audit, Contract, Estates, Hand Hygiene/IPC, Inc…

Shown above: Figure 14 [00:04:40] — Survey Management has its own Requirements (and Compliance Timeframes): Compliance Type "Feedback - SPC Flagge…

Step 7 [00:05:08 – 00:06:03]: SPC tables for F&F and CSAT

Action performed	Flicks through SPC sheets per service: "pulling a date down, pulling the formulas down".
Source → target	24 hr All Surveys → SPC sheets
State mutation	New month row added by hand (described)

Shown above: Figure 15 [00:05:21] — SPC - ND - F&F: monthly mean score per month (to 01/07/2026), mean / LCL / UCL, cumulative moving average and …

Shown above: Figure 16 [00:05:25] — SPC rule columns: Count > 3, Count < 3, Under Mean, 7 points < mean, Over Mean, 7 points > mean, Under LCL, Ov…

Shown above: Figure 17 [00:05:48] — CSAT sheets (VAS, Self Pay, NHS, All) follow the same SPC layout; tab list ends with Self Pay Types

Shown above: Figure 18 [00:06:01] — Self Pay Types: service description → code (ASD and ADHD → ND, Ear Care → EAR, No Scalpel Vasectomy → VAS, Joi…

Step 8 [00:06:03 – 00:06:20]: Compliance table

Action performed	"It builds up a compliance table here based on the statistical data." Selects F6, P5, R8.
Source → target	SPC sheets → Compliance
State mutation	—

Shown above: Figure 19 [00:06:07] — Formula bar, F6 Notes: SUBSTITUTE(…CONCAT(IF(N6>0,"SPC flagged - Overall satisfaction - "&TEXT(MAXIFS('SPC - V…

Shown above: Figure 20 [00:06:14] — Formula bar, P5: =XLOOKUP(EDATE(DATE(YEAR(TODAY()),MONTH(TODAY()),1),-1),'SPC - Self Pay - CSAT'!B:B,'SPC - Se…

Shown above: Figure 21 [00:06:17] — Formula bar, R8: =XLOOKUP(EDATE(DATE(YEAR(TODAY()),MONTH(TODAY()),1),0),'Z:\Performance Dashboards and Data\Us…

Step 9 [00:06:20 – 00:07:07]: Paste into the data store, last updated, save, refresh twice (described)

Action performed	Opens Compliance Data (UNABLE TO REFRESH), bottom of Compliance Data, Last Updated (People-Feedback 28/07/2026). "Bob's your uncle."
Source → target	Compliance → Compliance Data (paste); Last Updated
State mutation	Not performed

Shown above: Figure 22 [00:06:36] — Data store requirements (UNABLE TO REFRESH banner): Feedback - SPC Flagged - All / NHS / Self Pay / VAS / CTS …

Shown above: Figure 23 [00:06:42] — Compliance Data last row 4268: where the six Compliance rows are pasted (paste not performed on screen)

Shown above: Figure 24 [00:06:46] — Last Updated: People-Feedback updated 28/07/2026

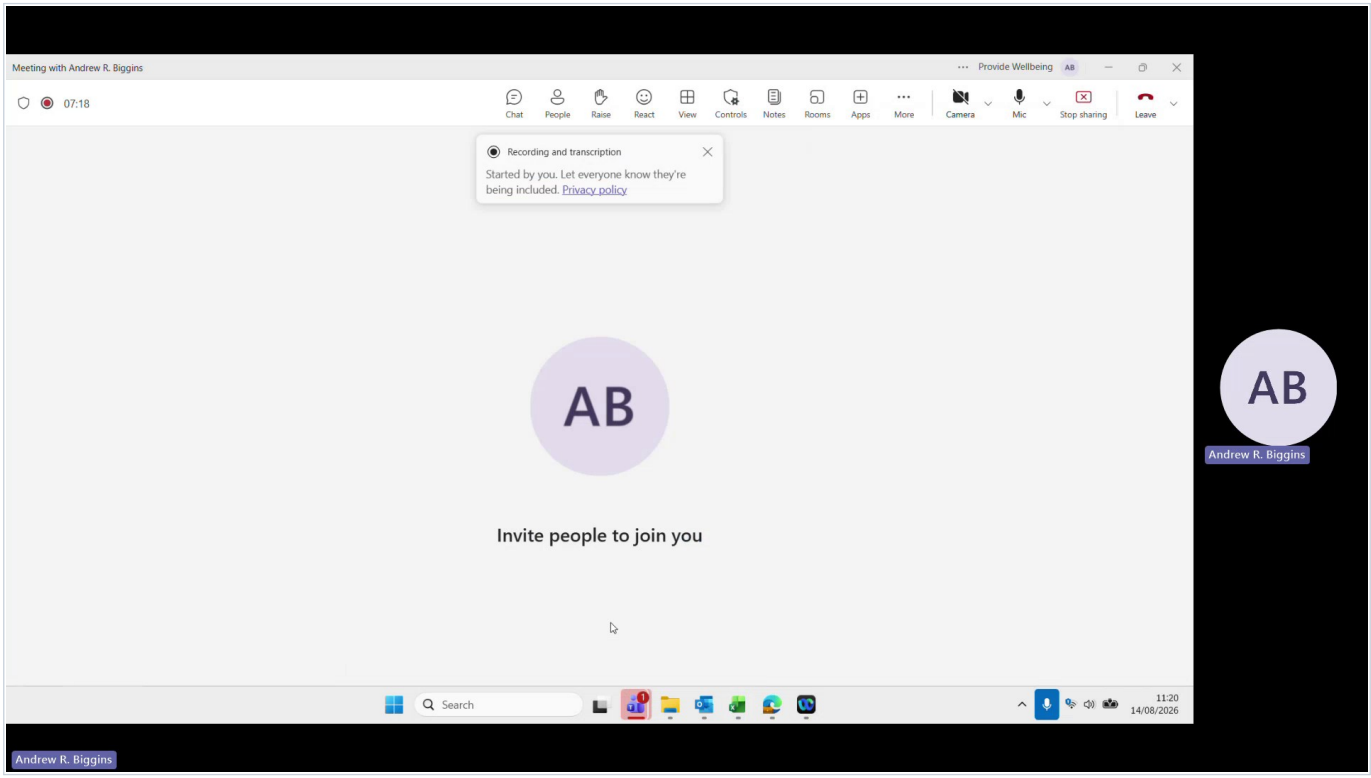


Figure 25 [00:07:05] — Teams recording › (end): Screen share stopped (Teams meeting window); recording ends at 07:07

4. Data Quality Checks, Manual Exceptions & Edge Cases

4.1 Business rules

ID	Rule	Evidence
BR-01	Responses without answers (“spoilt”) are excluded	Said 00:01:38
BR-02	Scores are aggregated to monthly averages per service (ND, CTS, VAS, Self Pay, NHS, All)	Figure 15 [00:05:21]
BR-03	A month is flagged when any SPC rule fires (3σ limits, 7-point runs and trends, 2 of 3 near a limit)	Figure 16 [00:05:25]
BR-04	Compliant = no flag last month for both CSAT and F&F	Figure 19 [00:06:07]
BR-05	Self Pay responses are classified by service via Self Pay Types	Figure 18 [00:06:01]

4.2 Spoken vs visual discrepancies and defects

ID	Said / intended	Seen	Impact
SV-01	“Four and five is acceptable, lower than four is not”	Sheets use monthly averages and SPC limits, not a per-response threshold	Acceptance rule unclear
SV-02	F&F: “less than five is not” acceptable	Same SPC logic as CSAT	Possibly mis-spoken; must be confirmed
SV-03	Two tools both needed?	Both build the same table; type names differ; both sets in the data store	Duplicate requirements in the store
SV-04	Refresh pulls the surveys	Self Pay query warning; combined query error during refresh	Silent partial data
FD-01	Last month	EDATE(…TODAY()…,-1)	Results depend on the day the file is opened
FD-02	Independent tool	Formula reads Incident Management.xlsm by path	Breaks if that file moves or is open/locked
FD-03	Clean column set	Stray “amp;” column from the export	Header drift in source files
FD-04	Usual constants	0.5 months, Missing TRUE, Issue 10, Unique Lookup with trailing dash	As in specs 01–04

4.3 Hidden failure modes

ID	Failure mode	Severity
FM-01	**Manual export and cleaning**: wrong date range or deleted valid responses	High
FM-02	**Hash-named files**: easy to save the wrong export over another	Medium
FM-03	**Power Query errors** after Survicate column changes	High
FM-04	**Duplicate tools and type names**	Medium
FM-05	**Hand-extended SPC ranges** (“pulling formulas down”)	Medium
FM-06	**Cross-workbook dependency** on Incident Management	Medium
FM-07	**Patient comments in shared spreadsheets**	High (IG)

5. Target Modern System Architecture Blueprint

Feedback runs on the shared platform (spec 01 §5) with a survey connector and an SPC rules engine.

5.1 Ingestion

- **Survicate API** (or scheduled export) per survey; responses stored once by Response ID (idempotent).
- **Spoilt-response rule** as code (e.g. no score answered), with counts shown for review instead of manual deletion.
- Comments stored in a restricted table, or not at all if the dashboard does not need them.

5.2 Data model

Table	Key columns
survey	survey_id, name, service_group (NHS / Self Pay), survicate_id
survey_response	response_id (PK), survey_id, responded_at, csat, ff, service_code, comment (restricted), is_spoilt
ref.self_pay_service_map	description → service_code (versioned)
ref.spc_rule (versioned)	rule_code, parameters (σ multiple, run length, window), applies_to
mart.feedback_monthly	month, service_code, measure (CSAT/FF), n, mean
mart.feedback_spc_signal	month, service_code, measure, rule_code, fired

5.3 SPC as SQL

```
WITH m AS (SELECT service_code, measure, month, avg(score) AS mean_score
           FROM mart.feedback_long GROUP BY 1,2,3),
lim AS (SELECT service_code, measure, avg(mean_score) AS cl, stddev_samp(mean_score) AS sd
        FROM m WHERE month < :as_of_month GROUP BY 1,2)
SELECT m.*, (m.mean_score < l.cl - 3*l.sd) AS under_lcl,
       (m.mean_score > l.cl + 3*l.sd) AS over_ucl,
       bool_and(m.mean_score < l.cl) OVER w7 AS seven_below_mean
FROM m JOIN lim l USING (service_code, measure)
WINDOW w7 AS (PARTITION BY m.service_code, m.measure ORDER BY m.month ROWS 6 PRECEDING);
```

- Baseline period, σ method and rule list are versioned parameters approved by the business.
- The compliance row per service is generated for the as-of month; Notes list fired rules, exactly like the current F6 text.
- One tool, one set of type names (resolves SV-03); no cross-workbook links (FD-02).

6. Gap & Ambiguity Register

ID	Timestamp	Gap / ambiguity	Ask
G-01	00:04:45	Is the first tool (24 hr All Surveys) still needed?	Decide; retire one
G-02	00:01:18	Exact rule for a “spoilt” response	Define
G-03	00:00:59–00:01:09	Acceptance threshold per response vs SPC on averages; F&F “less than five”	Confirm the rule
G-04	00:05:19	SPC baseline period, σ method, which rules count	Share Matt’s design or approve defaults
G-05	00:06:17	Why Compliance reads Incident Management Estates	Explain or remove
G-06	00:03:19	Cause of the Power Query warning/error	Share M code
G-07	00:00:47	Which Survicate surveys count (NHS VAS & POD includes podiatry?)	Confirm scope
G-08	whole video	Whether the dashboard shows comments or only compliance	Share measures

7. What Is Required to Build the Modern System

Feedback-specific requirements with reasons; shared platform items are in spec 01 §7. Client checklist: **Feedback_PreBuild_Client_Requirements.pdf**.

7.1 Functional requirements

ID	Requirement (priority)	Why it is required
FR-F01	Survicate connector with idempotent response load (M)	Manual export and hash-named files (FM-01, FM-02, Figure 4 [00:01:24]).
FR-F02	Coded spoilt-response rule with review counts (M)	Manual deletion today (BR-01).
FR-F03	Versioned self-pay service map (M)	Figure 18 [00:06:01].
FR-F04	SPC engine with versioned parameters (M)	Replaces hand-extended SPC sheets (FM-05, Figure 16 [00:05:25]).
FR-F05	Generated compliance rows with rule-hit notes (M)	Replaces the Compliance tab (Figure 19 [00:06:07]).
FR-F06	Single tool and naming (M)	Two overlapping tools (SV-03).
FR-F07	SPC charts for review (S)	The SPC sheets are the operator’s evidence.

7.2 Non-functional and testing

ID	Requirement	Why it is required
NFR-F01	Comments restricted or not stored	Patient free text (FM-07).
NFR-F02	Schema-drift alerts on survey exports	Query errors after column changes (FM-03, Figure 10 [00:03:22]).
T-F01	Recompute one month and match the Compliance tab rows (all six “within SPC criteria” on 14/08/2026)	Parity.
T-F02	Rule unit tests for each SPC rule	Core logic.

Appendix A. Figure Index

Fig	Time	File › Tab	Shows
1	00:00:05	Teams recording › (camera off)	Recording starts: “this video is how we update our survey feedback compliance”
2	00:00:48	Survicate (web) › Shared surveys	Survicate survey platform (panel.survicate.com), opened from the Provide intranet bookmarks: folders NHS ASD (1), NHS CTS (3), NHS VAS & POD (3), ...
3	00:01:15	Survicate (web) › NHS ASD/ADHD Post-Assessment (24 hours)	Survey results: 1,282 responses since 30 Nov 2022, 91.05% completion rate, monthly response chart; Export button top right
4	00:01:24	File Explorer › Z:\Performance Dashboards and Data\Data Store\Surveys	The Surveys folder: 24 hr All Surveys plus four exports with hash-like names (50f982bbbd00319b, 8f3115757b582202, a16d7414468e60c1, 92797e16bc759d36),...
5	00:01:36	50f982bbbd00319b.xlsx › Survey report	Raw Survicate export: Date & Time, Response ID, Respondent, Device, Platform, Page, Q#1 (overall satisfaction), Q#2 (friends and family), Q#3, fre...
6	00:02:21	Survey Management.xlsm › Compliance	Survey Management.xlsm opens on Compliance: six rows (Feedback - SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND), completed 14/08/2026, Issue 10,...
7	00:02:30	24 hr All Surveys.xlsx › 24 hr All Surveys	24 hr All Surveys table after refresh: Date & Time, Overall Satisfaction (1–5), Feedback (blurred), Friends and Family (1–5), “amp;” column, Servi...
8	00:03:01	24 hr All Surveys.xlsx › SPC - ND - F&F	SPC sheet in the first tool: monthly averages from 01/09/2024 and the rule columns; N19 =IF(AND(N18=1,M19=1),IF(SUM(M19:M25)>=7,1,0),0) (7 points b...
9	00:03:18	24 hr All Surveys.xlsx › Power Query editor	Queries: 24 hr Self Pay Survey (warning icon), 24 hr ND/VAS/CTS Survey, Self_Pay_Types, 24 hr All Surveys = Table.Combine({CTS, Self Pay, ND, VAS}) (e...
10	00:03:22	24 hr All Surveys.xlsx › Power Query editor	24 hr Self Pay Survey query (warning icon) with 11 applied steps (Source → Navigation → Promoted Headers → Changed Type → Removed Other Columns → Rena...
11	00:04:05	24 hr All Surveys.xlsx › Requirements	Requirements in the first tool: Provide Wellbeing × SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND; Required TRUE, Core, Unique Lookup, Active TR...
12	00:04:16	24 hr All Surveys.xlsx › Compliance Timeframes	Compliance Timeframes in the first tool: H4 Unique Lookup =CONCAT(Group 1,"-",Group 2,"-",Group 3) → “People-Feedback-”; Every x m...
13	00:04:27	File Explorer › Z:\Performance Dashboards and Data\User input	User input folder: Survey Management.xlsm (second tool) beside Audit, Contract, Estates, Hand Hygiene/IPC, Incident, Medical Devices, Policy and Proce...
14	00:04:40	Survey Management.xlsm › Requirements	Survey Management has its own Requirements (and Compliance Timeframes): Compliance Type “Feedback - SPC Flagged - All” etc., prefixed “Feedback -” unl...
15	00:05:21	Survey Management.xlsm › SPC - ND - F&F	SPC - ND - F&F: monthly mean score per month (to 01/07/2026), mean / LCL / UCL, cumulative moving average and the rule flags; AC5 LCL =AC3-(AC4*3)
16	00:05:25	Survey Management.xlsm › SPC - Self Pay - F&F	SPC rule columns: Count > 3, Count < 3, Under Mean, 7 points < mean, Over Mean, 7 points > mean, Under LCL, Over UCL, Trend Up, 7 points T...
17	00:05:48	Survey Management.xlsm › SPC - VAS - CSAT	CSAT sheets (VAS, Self Pay, NHS, All) follow the same SPC layout; tab list ends with Self Pay Types
18	00:06:01	Survey Management.xlsm › Self Pay Types	Self Pay Types: service description → code (ASD and ADHD → ND, Ear Care → EAR, No Scalpel Vasectomy → VAS, Joint Injections → JNT INJ, Minor Skin Surg...
19	00:06:07	Survey Management.xlsm › Compliance	Formula bar, F6 Notes: SUBSTITUTE(...CONCAT(IF(N6>0,"SPC flagged - Overall satisfaction - "&TEXT(MAXIFS('SPC - VAS - CSAT'!B:...
20	00:06:14	Survey Management.xlsm › Compliance	Formula bar, P5: =XLOOKUP(EDATE(DATE(YEAR(TODAY()),MONTH(TODAY()),1),-1),'SPC - Self Pay - CSAT'!B:B,'SPC - Self Pay - CSAT'!C:C,&...
21	00:06:17	Survey Management.xlsm › Compliance	Formula bar, R8: =XLOOKUP(EDATE(DATE(YEAR(TODAY()),MONTH(TODAY()),1),0),'Z:\Performance Dashboards and Data\User input\[Incident Management.xlsm]...
22	00:06:36	Compliance Data.xlsm › requirements_Final_Self_Ref	Data store requirements (UNABLE TO REFRESH banner): Feedback - SPC Flagged - All / NHS / Self Pay / VAS / CTS / ND rows next to staff rows
23	00:06:42	Compliance Data.xlsm › Compliance Data	Compliance Data last row 4268: where the six Compliance rows are pasted (paste not performed on screen)
24	00:06:46	Compliance Data.xlsm › Last Updated	Last Updated: People-Feedback updated 28/07/2026
25	00:07:05	Teams recording › (end)	Screen share stopped (Teams meeting window); recording ends at 07:07

Appendix B. Key timestamped utterances

Time	Utterance (lightly cleaned)
00:00:05	This video is how we update our survey feedback compliance.
00:00:23	The first thing we do is download our surveys from the survey platform.
00:00:47	There are four surveys... 24-hour surveys.

Time	Utterance (lightly cleaned)
00:00:59	Four and five is acceptable, lower than four is not.
00:01:06	Four and five is acceptable, less than five is not.
00:01:18	We usually do a bit of a cleanup of the data.
00:01:46	We get rid of any spoilt ones.
00:02:16	This was designed by Matt, so we just follow his process.
00:03:19	It’s having trouble finding something with that one.
00:04:09	This is a bit confusing.
00:04:52	I’m not even sure we need to complete the first one.
00:05:47	It’s a case of pulling a date down, pulling the formulas down.
00:06:26	We copy that into the compliance data, into the bottom here.
00:06:50	We save it, refresh it twice.